MIL-M-28787/288A(EC)
28 April 1980
SUPERSEDING
MIL-M-28787/288(EC)
21 March 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CLOCK, MICROPROCESSOR, DIGITAL, KEY CODE JRH

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

* 1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic microprocessor clock module.

1.2 Classification. This specification covers two classes of modules;

class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.351 failures per million operating hours for class I or 0.518 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JRH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/288-1 for class I and M28787/288-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc and V_{DD} of 12.0 V ±10 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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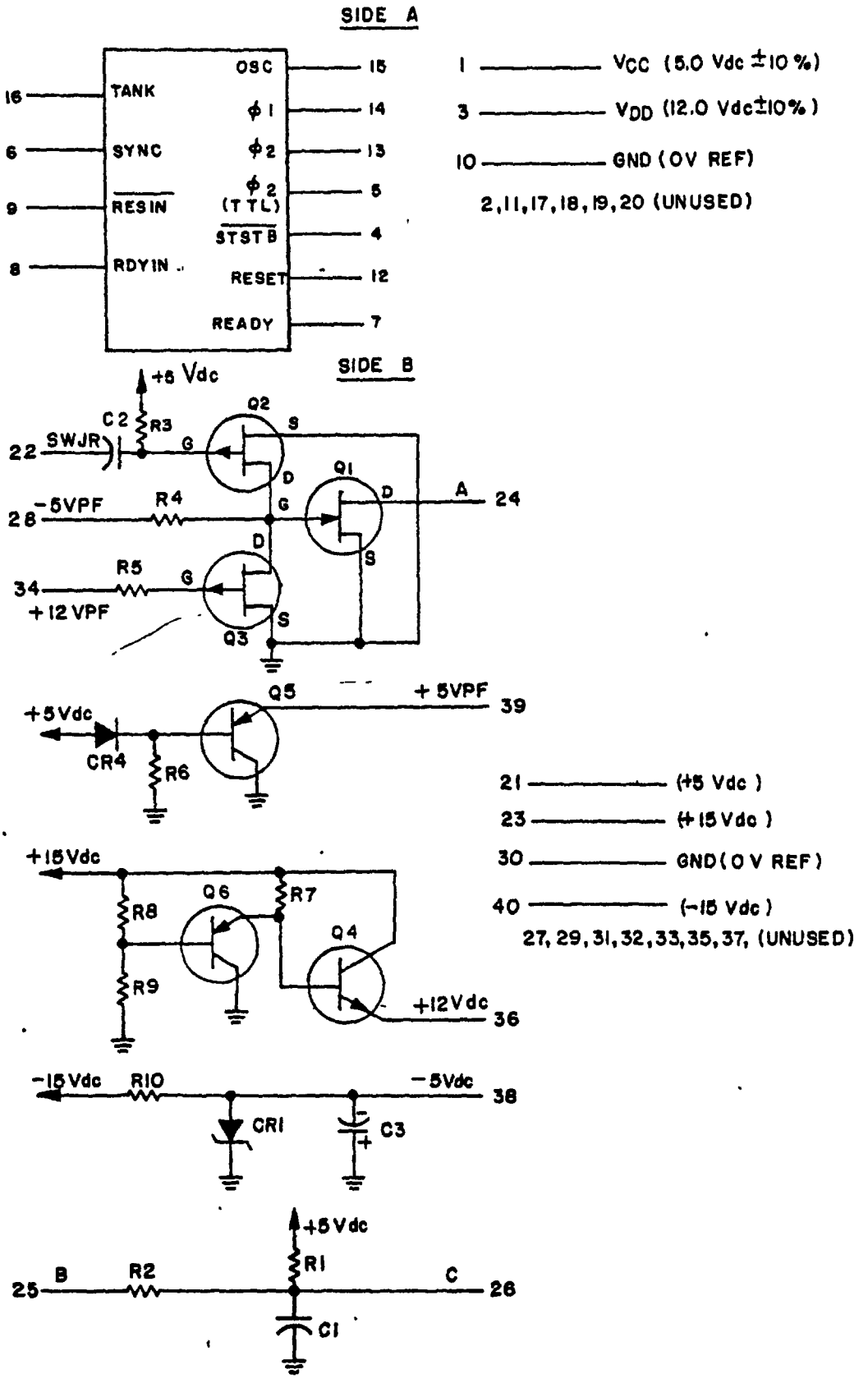


FIGURE 1. JRH module function diagram, microprocessor clock.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	Isolation (initial)	II	S	10.0	∞	10.0	∞	10.0	∞	10.0	∞	10.0	∞	MΩ
	(end-of-life)	II	S	1.0	∞	1.0	∞	1.0	∞	1.0	∞	1.0	∞	MΩ
B	V_{CC} and V_{DD} maximum stress	III	S	← Module must pass remaining tests →										
C1-C3	Output short circuit current	IV	X	-10	-60	-10	-60	-10	-60	-10	-60	-10	-60	mA
D1-D3	Input clamp diode voltage	V	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
E1-E3	Static output voltage, low level	VI	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
F1,F4,F5	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F2,F3	Static output voltage, high level	VII	X	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	V
F6,F7	Static output voltage, high level	VII	X	9.0	10.8	9.0	10.8	9.0	10.8	9.0	10.8	9.0	10.8	V
G1,G3,G5	Power fail/restart voltage, low level	VIII	X	0.0	0 8	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
G7	Power fail/restart voltage, low level	VIII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
G2,G4,G6, G8	Power fail/restart voltage, high level	VIII	X	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0	V
H1,H2,H5	-5 V regulator voltage	IX	X	-4.85	-5.35	-4.80	-5.35	-4.85	-5.41	-4.72	-5.35	-4.85	-5.47	V
H3,H4	12 V regulator voltage	IX	X	11.4	12.6	11.1	12.6	11.4	12.92	10.67	12.6	11.4	13.29	V
H6	12 V regulator voltage	IX	X	11.4	12.6	11 1	12.6	11.4	12.92	10.67	12.6	11.4	13.29	V
J1	-5 V regulator ripple rejection	X	X	0	10	0	10	0	10	0	10	0	10	mVrms
J2	12 V regulator ripple rejection	X	X	0	24	0	24	0	24	0	24	0	24	mVrms
K	Output leakage current	XI	X	0	500	0	500	0	500	0	500	0	500	nA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L3	Input current, low level	XII	X	-1	-270	-1	-270	-1	-270	-1	-270	-1	-270	µA
M1-M3	Input current, high level	XII	X	0	10	0	10	0	10	0	10	0	10	µA
N1	V_{DD} power supply current	XIII	X	0.01	12.0	0.01	12.0	0.01	12.0	0.01	12.0	0.01	12.0	mA
N2	V_{CC} power supply current	XIII	X	0.01	115	0.01	115	0.01	115	0.01	115	0.01	115	mA
P1,P2	Transition time (t_{TLH}), (ϕ1 and ϕ2), C_L = 50 pF	XIV	S	1	25	1	25	1	28	1	22	1	31	ns
P3-P7	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	1	15	1	15	1	17	1	13	1	19	ns
Q1,Q2	Transition time (t_{THL}), (ϕ1 and ϕ2), C_L = 50 pF	XV	S	1	25	1	25	1	28	1	22	1	31	ns
Q3-Q7	Transition time (t_{THL}), C_L = 50 pF	XV	S	1	15	1	15	1	17	1	15	1	19	ns
R	ϕ1 pulse width check ($t_{\phi 1}$), C_L = 50 pF	XVI	S	90	115	90	115	90	115	90	115	90	115	ns
S	ϕ2 pulse width check ($t_{\phi 2}$), C_L = 50 pF	XVI	S	231	275	231	275	231	275	231	275	231	275	ns
T	ϕ1 to ϕ2 delay (t_{D1}), C_L = 50 pF	XVI	S	0	20	0	10	0	25	0	10	0	30	ns
U	ϕ2 to ϕ1 delay (t_{D2}), C_L = 50 pF	XVI	S	85	115	85	115	85	120	90	115	95	125	ns
V	ϕ1 to ϕ2 delay (t_{D3}), C_L = 50 pF	XVI	S	110	150	110	150	110	155	115	150	120	160	ns
W	ϕ2 to ϕ2 (TTL) delay (t_{DLH}), C_L= 50 pF	XVI	S	-5	+15	-5	+15	-5	+17	-3	+15	-1	+19	ns
X	ϕ2 to ϕ2 (TTL) delay (t_{DHL}), C_L = 50 pF	XVI	S	-5	+15	-5	+15	-5	+17	-3	+15	-1	+19	ns
Y	ϕ1 pulse period check (T_{CY}), C_L = 50 pF	XVI	S	495	505	495	505	495	505	495	505	495	505	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z	$\phi 2$ to $\overline{STSTB}$ delay (t_{DSS}), C_L = 50 pF	XVI	S	304	334	304	334	309	339	269	299	314	344	ns
AA	$\overline{STSTB}$ pulse width check (t_{PW}), C_L = 50 pF	XVI	S	32	60	32	60	32	60	32	60	32	60	ns
BB	Ready to $\phi 2$ delay (t_{PHL}), (t_{DR1}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
CC	Ready to $\phi 2$ delay (t_{PHL}), (t_{DR2}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
DD	Reset to $\phi 2$ delay (t_{PHL}), (t_{DR3}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
EE1	Frequency check, 18 MHz	XVII	X	17.9	18.1	17.9	18.1	17.9	18.1	17.9	18.1	17.9	18.1	MHz
EE2	Frequency check, 2 MHz	XVII	S	1.98	2.02	1.98	2.02	1.98	2.02	1.98	2.02	1.98	2.02	MHz
FF1, FF2	Power supply transient amplitude (V_{CC} and V_{DD})	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
GG1-GG3	Output crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
GG4,GG7	Output crosstalk, low level	XIX	Q	0	850	0	850	0	850	0	850	0	850	mV
GG5,GG6	Output crosstalk, low level	XIX	Q	0	1,000	0	1,000	0	1,000	0	1,000	0	1,000	mV
HH1-HH7	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
JJ1,JJ2	Power supply filter capacitance	XXI	S	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	uF
KK1-KK3	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through KK shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through KK in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through EE and GG through JJ shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions JJ1 and JJ2 shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through EE and GG through JJ shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through JJ shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module function. The module consists of a clock generator driver to be used in conjunction with the HRH module. The module also contains a -5 V and a +12 V regulator. These regulators should only be used with the JRH or the HRH modules. The module also has circuits to be used as power fail/restart for the HRH module if the HRH module supplies are connected to this circuitry.

* 6.4.1 Input to output relationship. By providing +5 V to pin 1 and +12 V to pin 3, the "A" side of the module will provide all necessary clocking signals for the HRH module. A stable crystal controlled 2 MHz clock out of the ϕ2 (TTL) output pin 5 can be provided for any system by providing 5 V to pin 1 and pin 3. The "B" side of the module provides a power fail/restart system when pins 22, 28, and 34 are attached to

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the HRH supplies and pin 24 is attached to pin 25. If any supply is disconnected or goes below ±1 0 V for the ±5 V supplies or +4 V for the +12 V supply, the appropriate FET turns on and begins to discharge C1 through R2. If this condition exists in excess of 500 µs, C1 is sufficiently discharged to look like a logic "L", when power is reapplied, capacitor C1 will charge through R1 in approximately 30 ms. The output at pin 26 can be wired to the Resin input pin 9 on the "A" side of this module, and the reset output pin 12 can be connected to a HRH module to restart it at location (0000) should a power supply fail. Pin 22 provides a differentiator circuit, resistor R3 and capacitor C2, which can be used as a software interrupt. Should pin 22 be held in a logic "L" state for more than 2 ms, the HRH module will be reset to location (0000) in the same manner as the power fail/restart. With pin 39 connected to pin 25, a power fail indicator is provided for systems using this module as a 2 MHz oscillator with only +5 V available. The output of the -5 V regulator is pin 38. It will provide 1.0 mA at -5 V. Also the output of the 12 V regulator is pin 36. It will provide 200 mA at 12 V. Both regulator outputs should only be used to drive the JRH or the HRH modules.

* 6.4.2 Parallel outputs. Paralleling of outputs is not permissible.

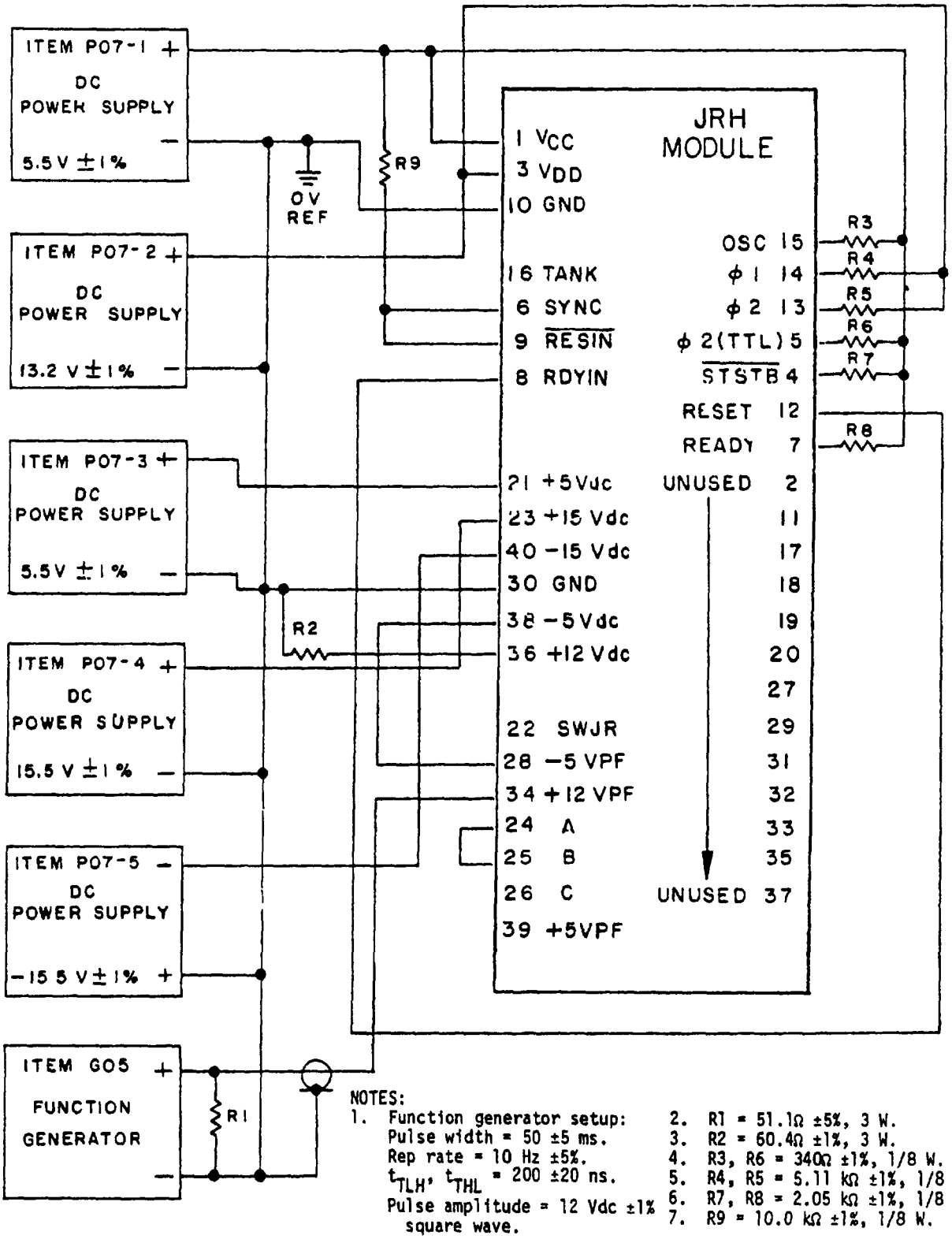
6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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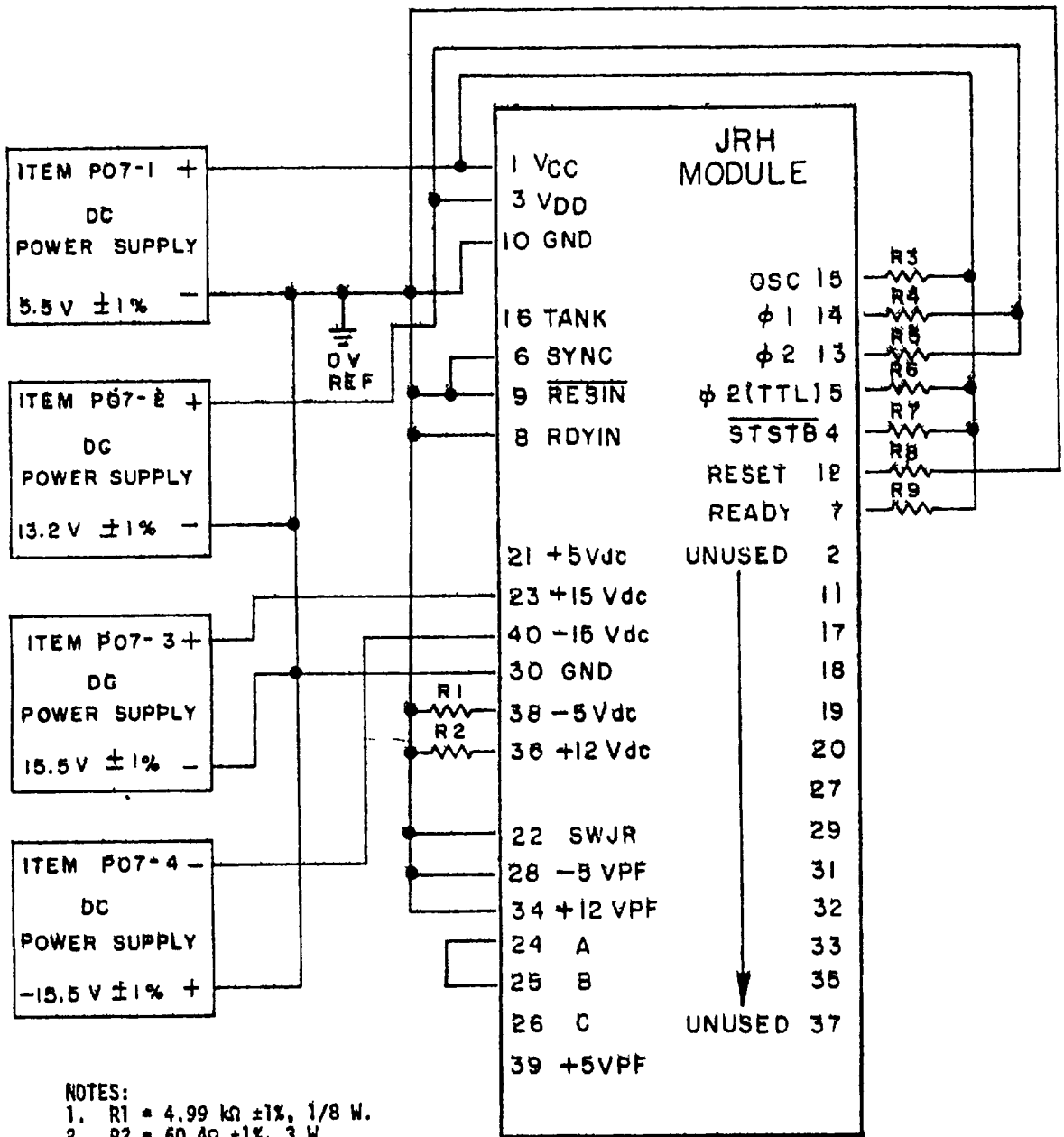
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NOTES:
1. Function generator setup: Pulse width = 50 ±5 ms. Rep rate = 10 Hz ±5%. t_{TLH}, t_{THL} = 200 ±20 ns. Pulse amplitude = 12 Vdc ±1% square wave.
2. R1 = 51.1Ω ±5%, 3 W.
3. R2 = 60.4Ω ±1%, 3 W.
4. R3, R6 = 340Ω ±1%, 1/8 W.
5. R4, R5 = 5.11 kΩ ±1%, 1/8 W.
6. R7, R8 = 2.05 kΩ ±1%, 1/8 W.
7. R9 = 10.0 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 4.99 kΩ ±1%, 1/8 W.
2. R2 = 60.4Ω ±1%, 3 W.
3. R3, R6 = 340Ω ±1%, 1/8 W.
4. R4, R5 = 5.11 kΩ ±1%, 1/8 W.
5. R7, R9 = 2.05 kΩ ±1%, 1/8 W.
6. R8 = 34.0 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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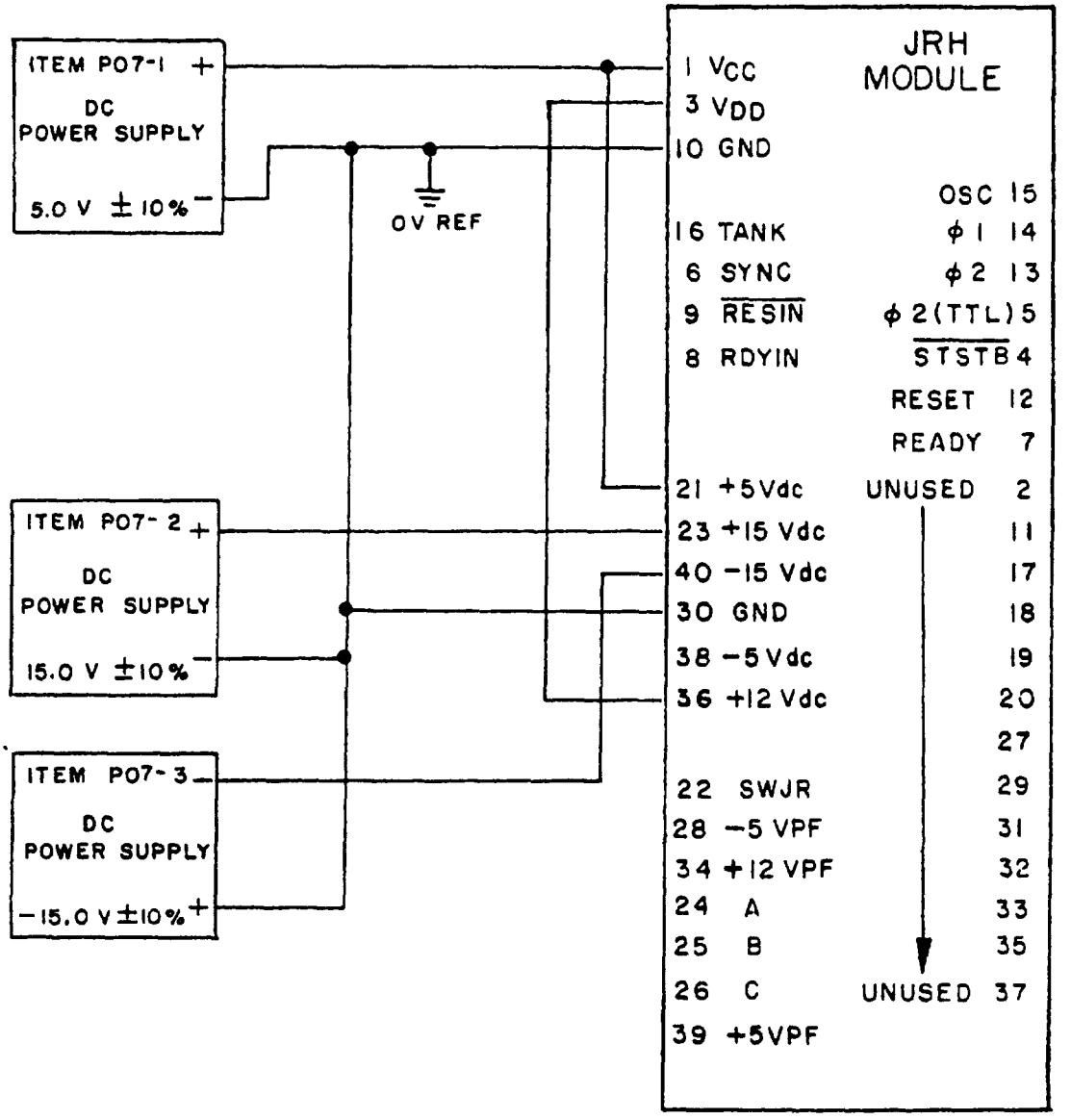


FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

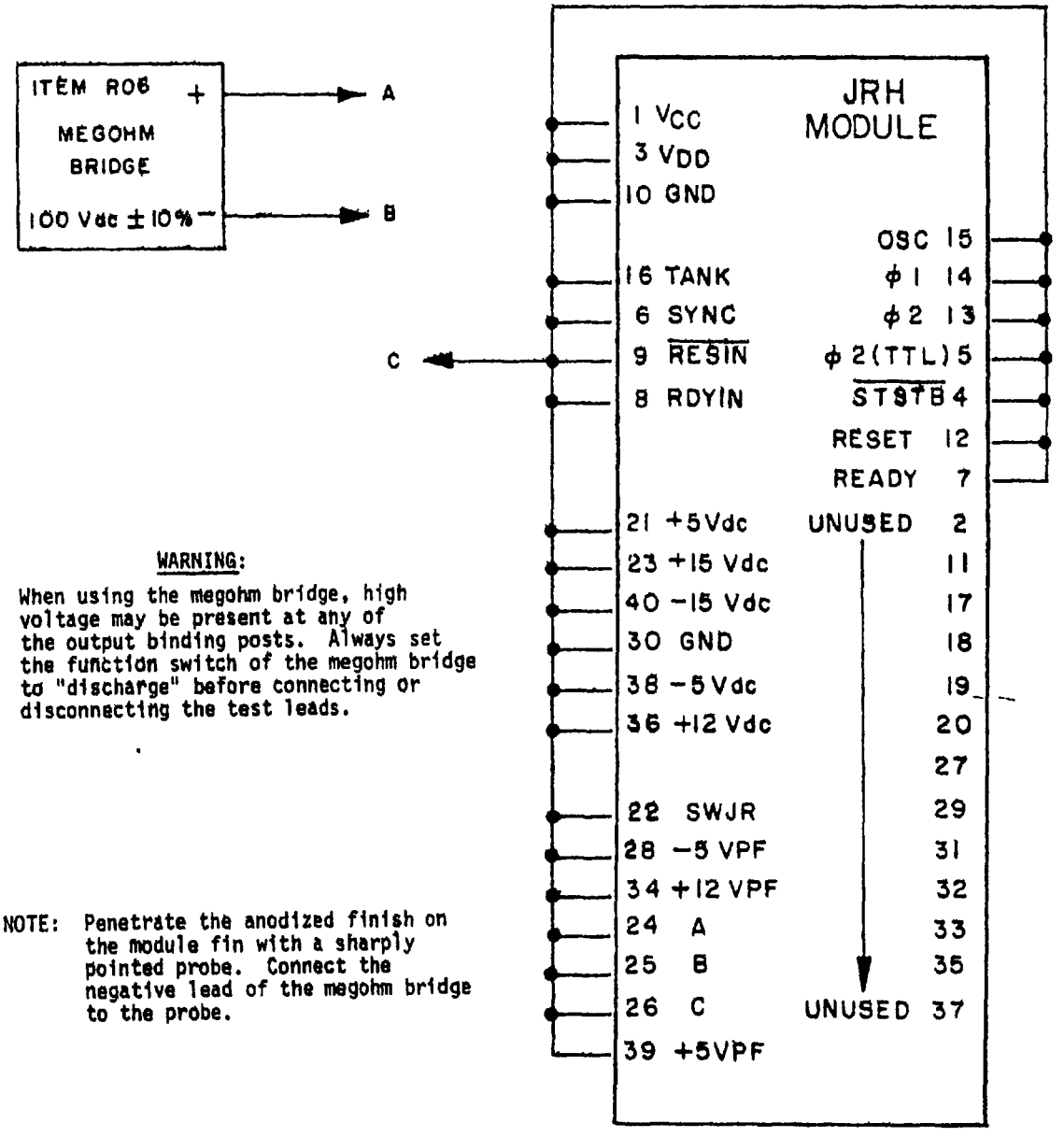
Test condition	Parameter	Connect		Initial limits +25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,2,11,17,18, 19,20,27,29,31, 32,33,35,37	Module frame	10.0	∞	MΩ	5
A2		2	11,17,18,19, 20,27,29,31, 32,33,35,37				
A3		11	2,17,18,19, 20,27,29,31, 32,33,35,37				
A4		17	2,11,18,19, 20,27,29,31, 32,33,35,37				
A5		18	2,11,17,19, 20,27,29,31, 32,33,35,37				
A6		19	2,11,17,18, 20,27,29,31, 32,33,35,37				
A7		20	2,11,17,18, 19,27,29,31, 32,33,35,37				
A8		27	2,11,17,18, 19,20,29,31, 32,33,35,37				
A9	Isolation	29	2,11,17,18, 19,20,27,31, 32,33,35,37	10.0	∞	MΩ	5

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TABLE II. Isolation test -continued.

Test condition	Parameter	Connect		Initial limits +25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A10	Isolation	31	2,11,17,18, 19,20,27,29, 32,33,35,37	10.0	∞	MΩ	5
A11		32	2,11,17,18, 19,20,27,29, 31,33,35,37				
A12		33	2,11,17,18, 19,20,27,29, 31,32,35,37				
A13		35	2,11,17,18, 19,20,27,29, 31,32,33,37				
A14		37	2,11,17,18, 19,20,27,29, 31,32,33,35				
* A15	Isolation	1	21	10.0	∞	MΩ	5

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module fin with a sharply pointed probe. Connect the negative lead of the megohm bridge to the probe.

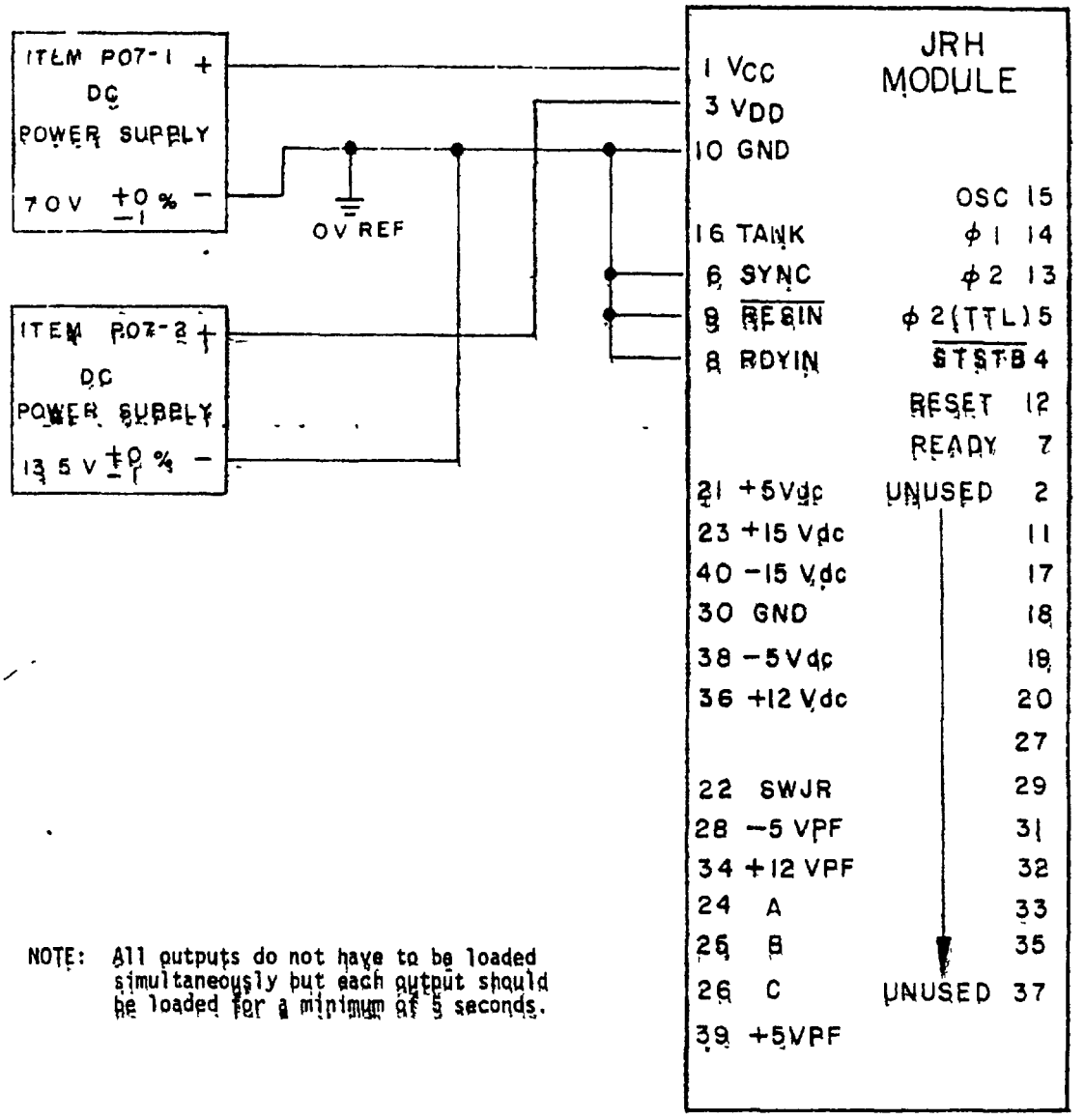
FIGURE 5. Isolation test circuit for table II.

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TABLE III. V_{CC} and V_{DD} maximum stress test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
B	V_{CC} and V_{DD} maximum stress	Module must pass remaining tests			6

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NOTE: All outputs do not have to be loaded simultaneously but each output should be loaded for a minimum of 5 seconds.

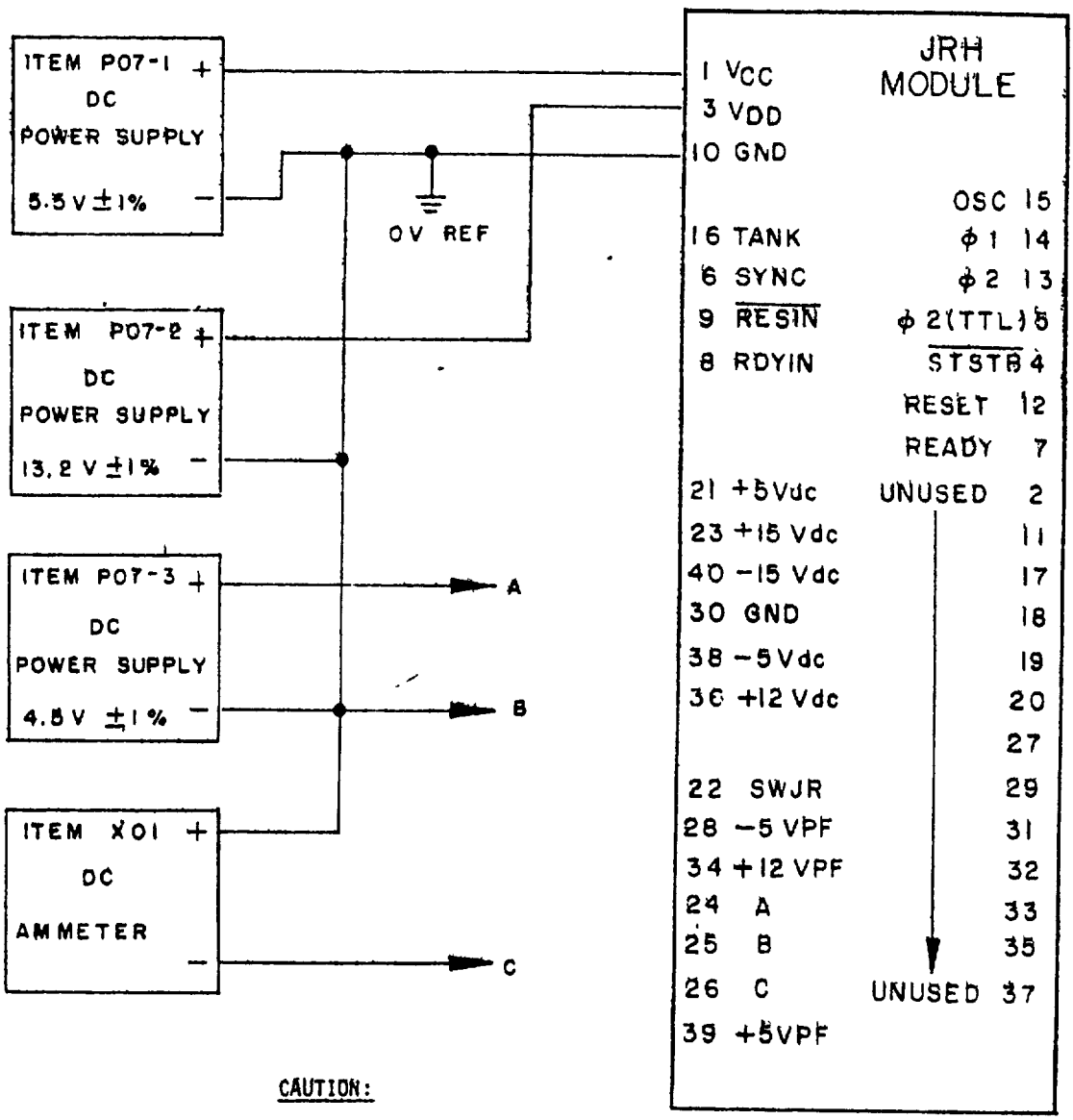
FIGURE 6. V_{CC} and V_{DD} maximum stress test circuit for table III.

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TABLE IV. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
C1	$\overline{\text{STSTB}}$	Output short circuit current	9	6,8	4	-10	-60	mA	7
C2	Reset	Output short circuit current	Open	6,9,8	12	-10	-60	mA	7
C3	Ready	Output short circuit current	8	6,9	7	-10	-60	mA	7

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Only one output shall be shorted at a time and the measurement taken within 5 seconds.

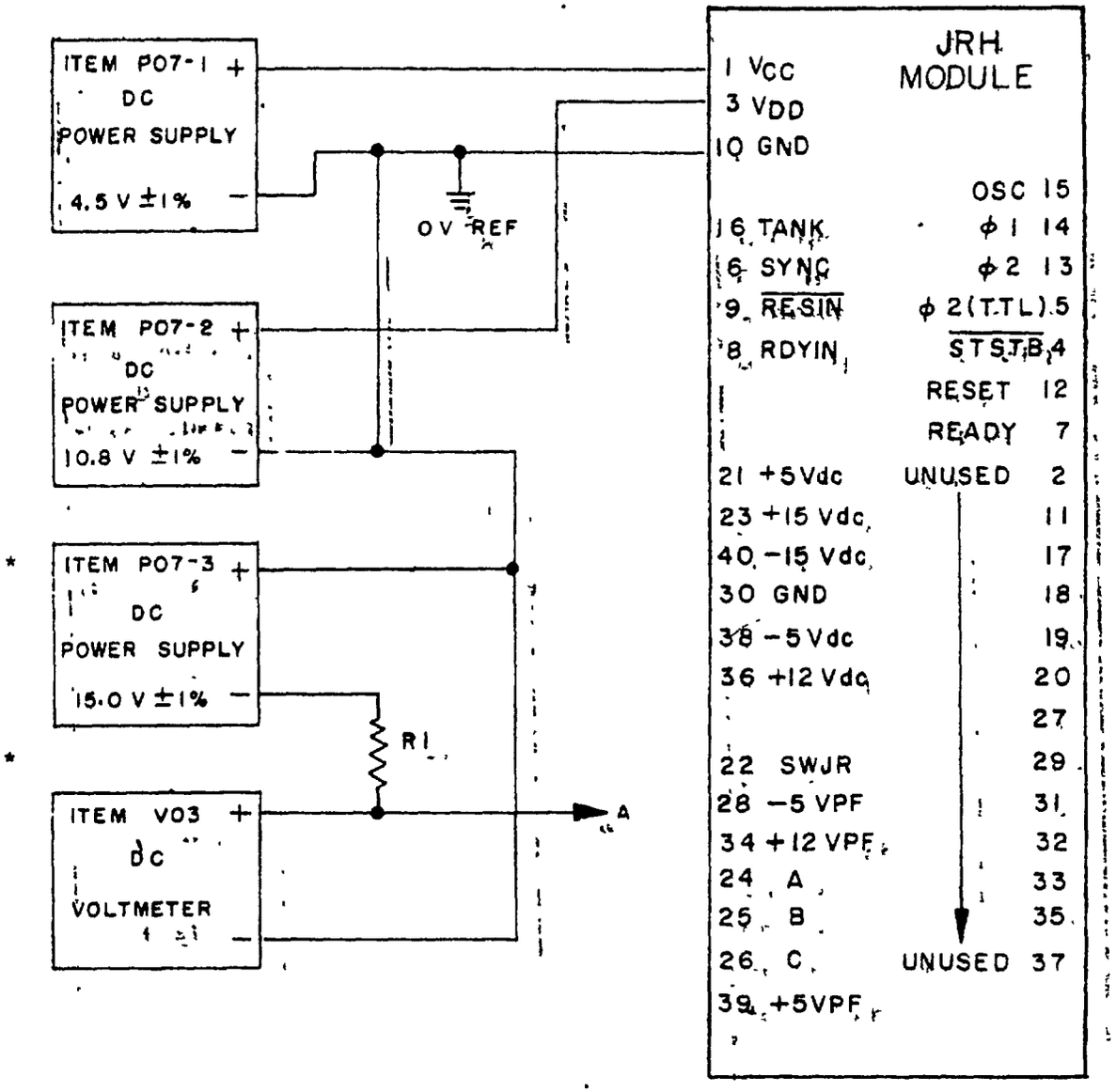
FIGURE 7. Output short circuit current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
D1	Sync	Input clamp diode voltage	6	-0.4	-1.2	V	8
D2	$\overline{\text{Resin}}$	Input clamp diode voltage	9	-0.4	-1.2	V	8
D3	RDYIN	Input clamp diode voltage	8	-0.4	-1.2	V	8

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NOTE: R = 2.74 kΩ ±1%, 1/8 W.

FIGURE 8. Input clamp diode voltage test circuit for table V.

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TABLE VI. Static output voltage (low-level) test.

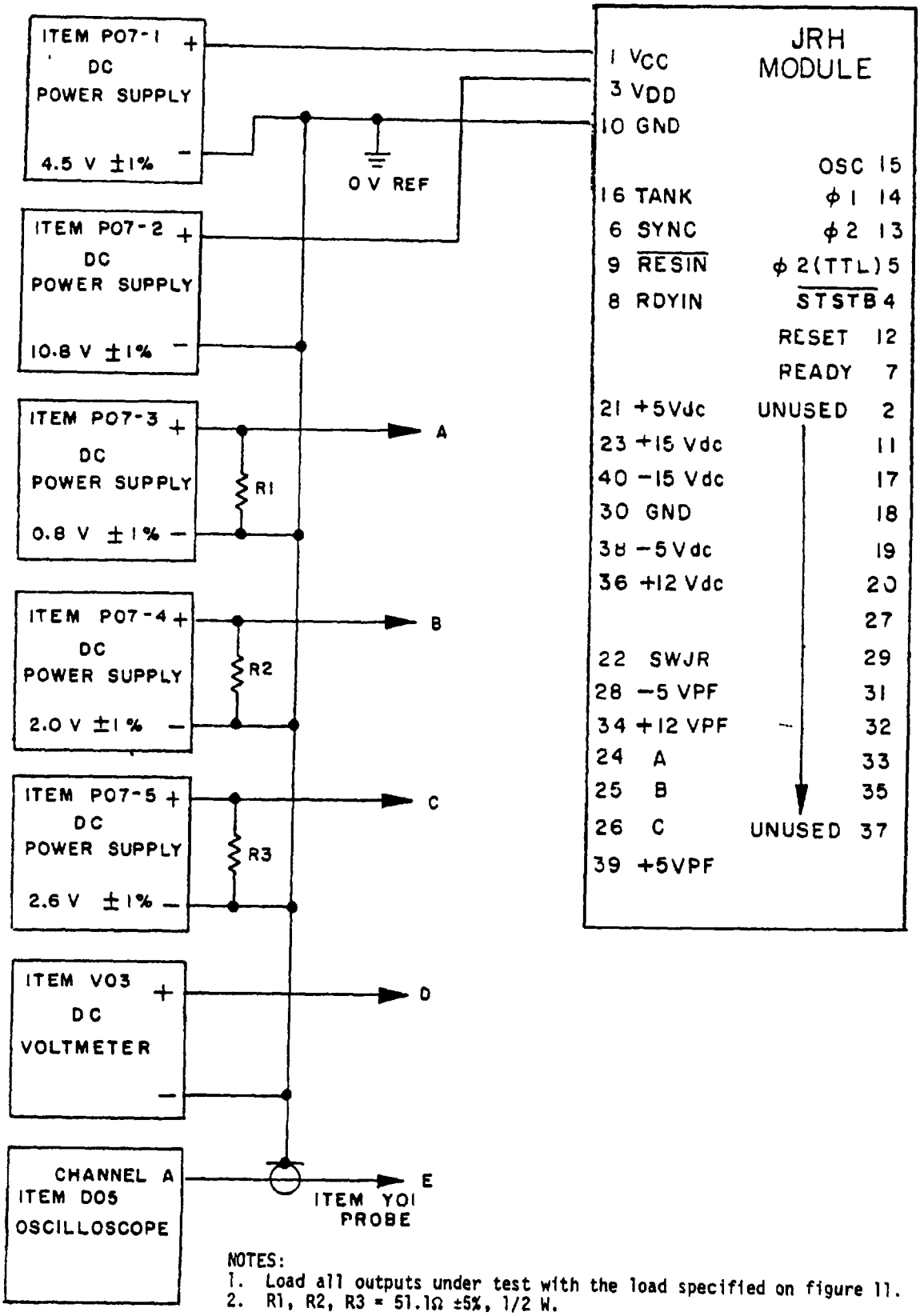
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	$\overline{\text{STSTB}}$ Static output voltage, low level	6,9	8	Open	4	Open	0.00	0.45	V	9
E2	Reset Static output voltage, low level	6	8	9	12	Open	0.00	0.45	V	9
E3	Ready Static output voltage, low level	6,8,9	Open	Open	7	Open	0.00	0.45	V	9

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TABLE VII. Static output voltage (high-level) test.

Test condition	Parameter		Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	$\overline{STSTB}$	Static output voltage, high level	6,8	Open	9	4	Open	2.4	4.5	V	9
F2	Reset		6,8,9	Open	Open	12	Open	3.3			9
F3	Ready		6	8	9	7	Open	3.3			9
F4	OSC		6,8,9	Open	Open	Open	15	2.4			9,10
F5	Φ2(TTL)						5	2.4	4.5		
F6	Φ1						14	9.0	10.8		
F7	Φ2	Static output voltage, high level	6,8,9	Open	Open	Open	13	9.0	10.8	V	9,10

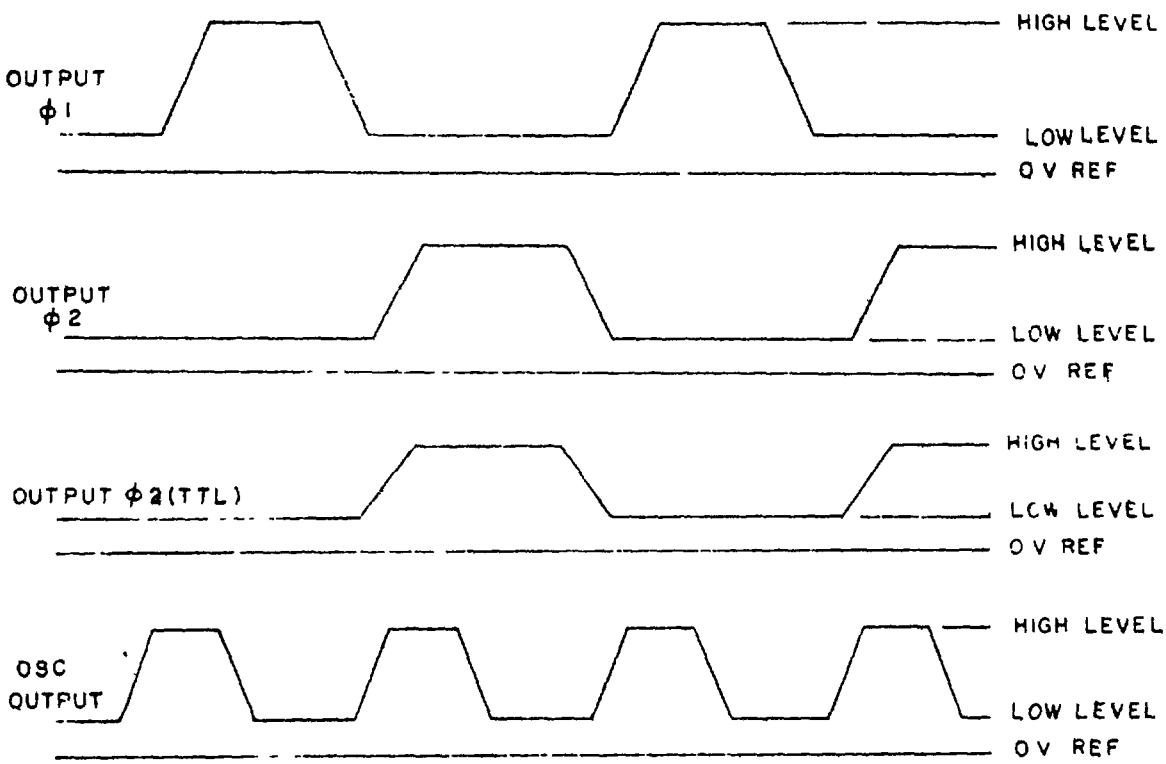
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NOTES:
1. Load all outputs under test with the load specified on figure 11.
2. R1, R2, R3 = 51.1Ω ±5%, 1/2 W.

FIGURE 9. Static output voltage (low-level and high-level) test circuit for tables VI and VII.

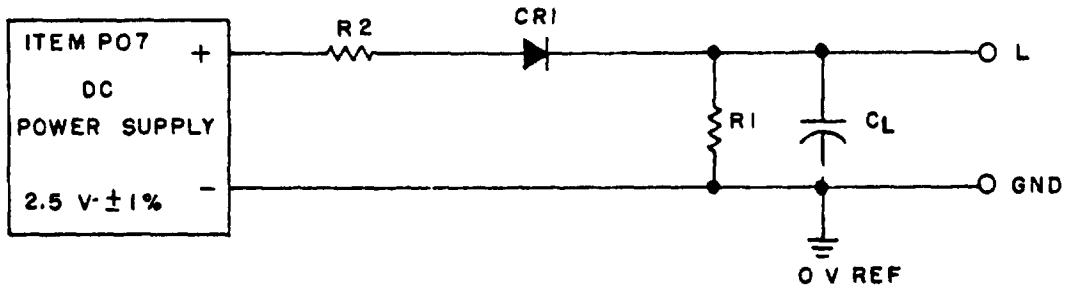
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* NOTE: When measuring high level voltage, measure between high level as indicated on the appropriate waveform and 0 V ref.

* FIGURE 10. Static output voltage (high-level) test circuit waveforms for table VII.

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NOTES:
1. Logic load for pins 5 and 15.

   a. R1 = 2.43 kΩ ±1%, 1/8 W.
   b. R2 = Select value for current flow of 10.0 mA ±5% out of point "L" with point "L" held at 0.45 V.
   c. CR1 = 1N4150,or equivalent.
   d. C_L = 50 pF ±5% (including probe and parasitic capacitance).
   e. During output switching, the 2.5 V power supply shall have a maximum overshoot of 250 mV.

2. Logic load for pin 4.

   a. R1 = 2.43 kΩ ±1%, 1/8 W.
   b. R2 = Select value for current flow of 2.5 mA ±5% out of point "L" with point "L" held at 0.45 V.
   c. CR1 = 1N4150,or equivalent.
   d. C_L = 50 pF ±5% (including probe and parasitic capacitance).
   e. During output switching, the 2.5 V power supply shall have a maximum overshoot of 250 mV.

3. Logic load for pins 7 and 12.

   a. R1 = 34.0 kΩ ±1%, 1/8 W.
   b. R2 = Select value for current flow of 2.5 mA ±5% out of point "L" with point "L" held at 0.45 V.
   c. CR1 = 1N4150,or equivalent.
   d. C_L = 50 pF ±5% (including probe and parasitic capacitance).
   e. During output switching, the 2.5 V power supply shall have a maximum overshoot of 250 mV.

4. Logic load for pins 13 and 14.

   a. R1 = 93.1 kΩ ±1%, 1/8 W.
   b. R2 = Select value for current flow of 2.5 mA ±5% out of point "L" with point "L" held at 0.45 V.
   c. CR1 = 1N4150,or equivalent.
   d. C_L = 50 pF ±5% (including probe and parasitic capacitance).
   e. During output switching, the 2.5 V power supply shall have a maximum overshoot of 250 mV.

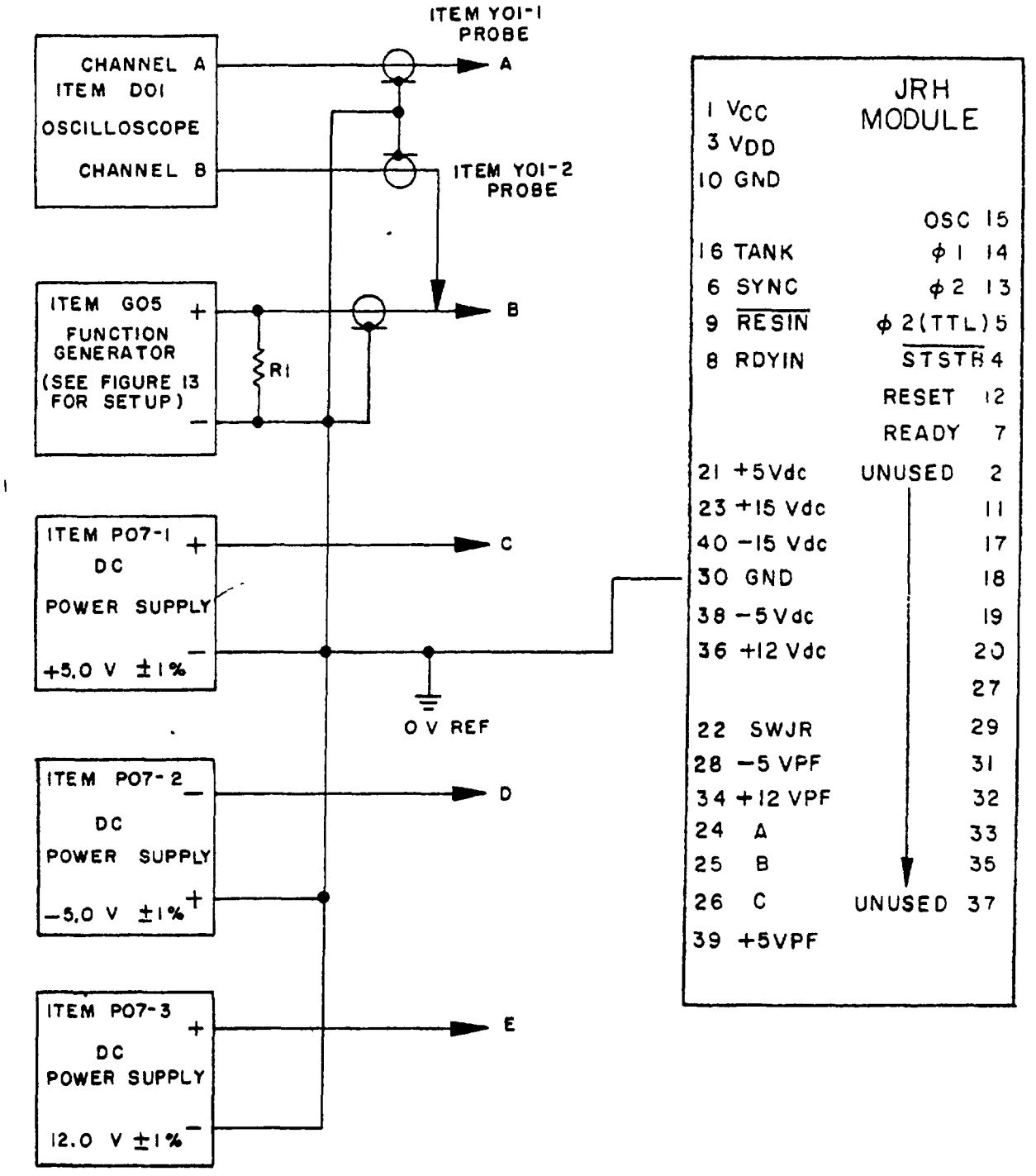
FIGURE 11. Logic load circuit.

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TABLE VIII. Power fail/restart voltage check.

Test condition	Parameter		Connect						Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Pin 25 to pin	Min	Max		
G1	-5 V PF (low level)	Power fail/restart voltage check	26	28	21	Open	34	24	0.0	0.8	V	12,13
G2	-5 V PF (high level)			28		Open	34		3.0	5.0		
G3	+12 V PF (low level)			34		28	Open		0.0	0.8		
G4	+12 V PF (high level)			34			Open		3.0	5.0		
G5	SWJR (low level)			22			34		0.0	0.8		
G6	SWJR (high level)			22	21	28	34	24	3.0	5.0		
G7	+5 V PF (low level)			21	Open	Open	Open	39	0.0	1.5		
G8	+5 V PF (high level)	Power fail/restart voltage check	26	21	Open	Open	Open	39	3.0	5.0	V	12,13

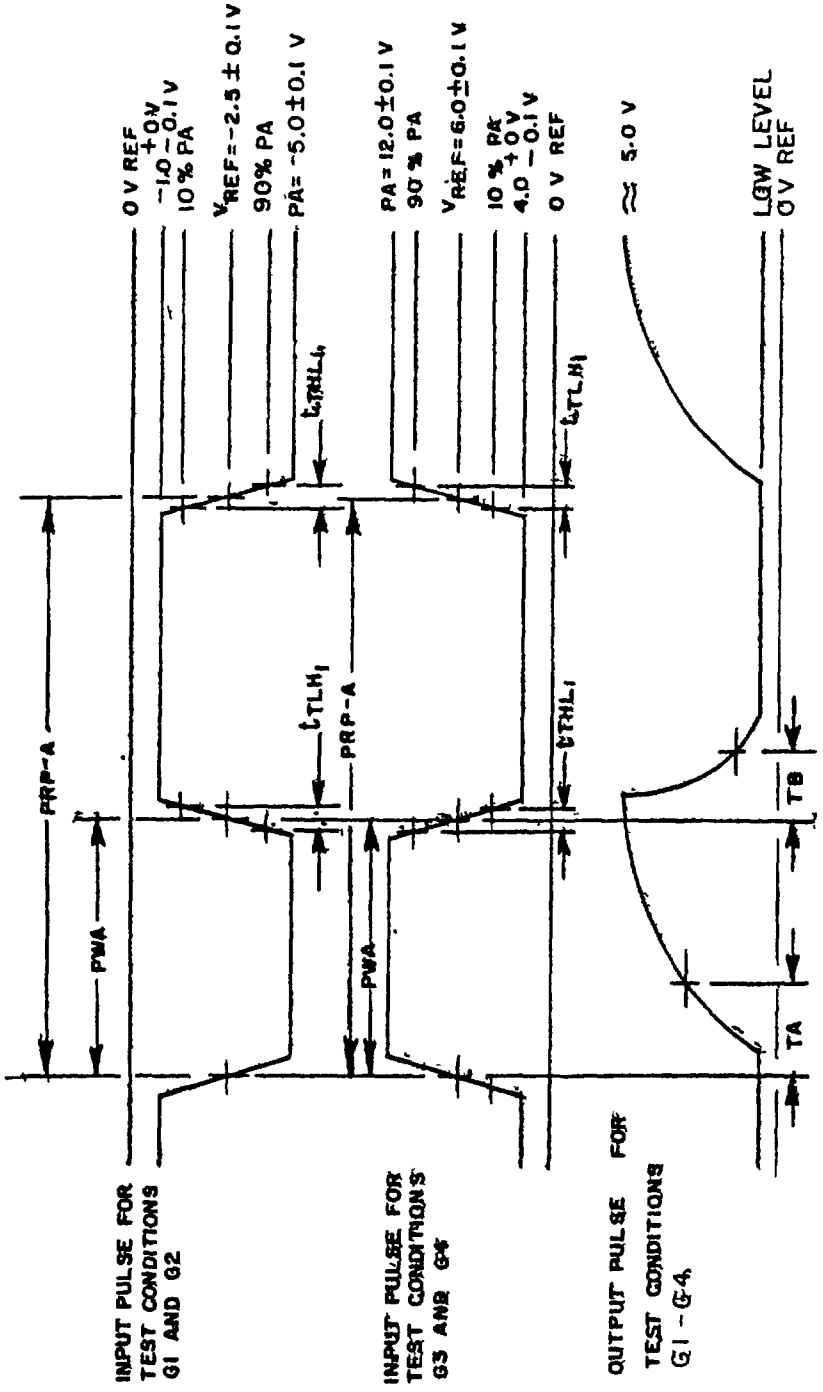
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NOTE: R1 = 51.1Ω ±5%, 3 W.

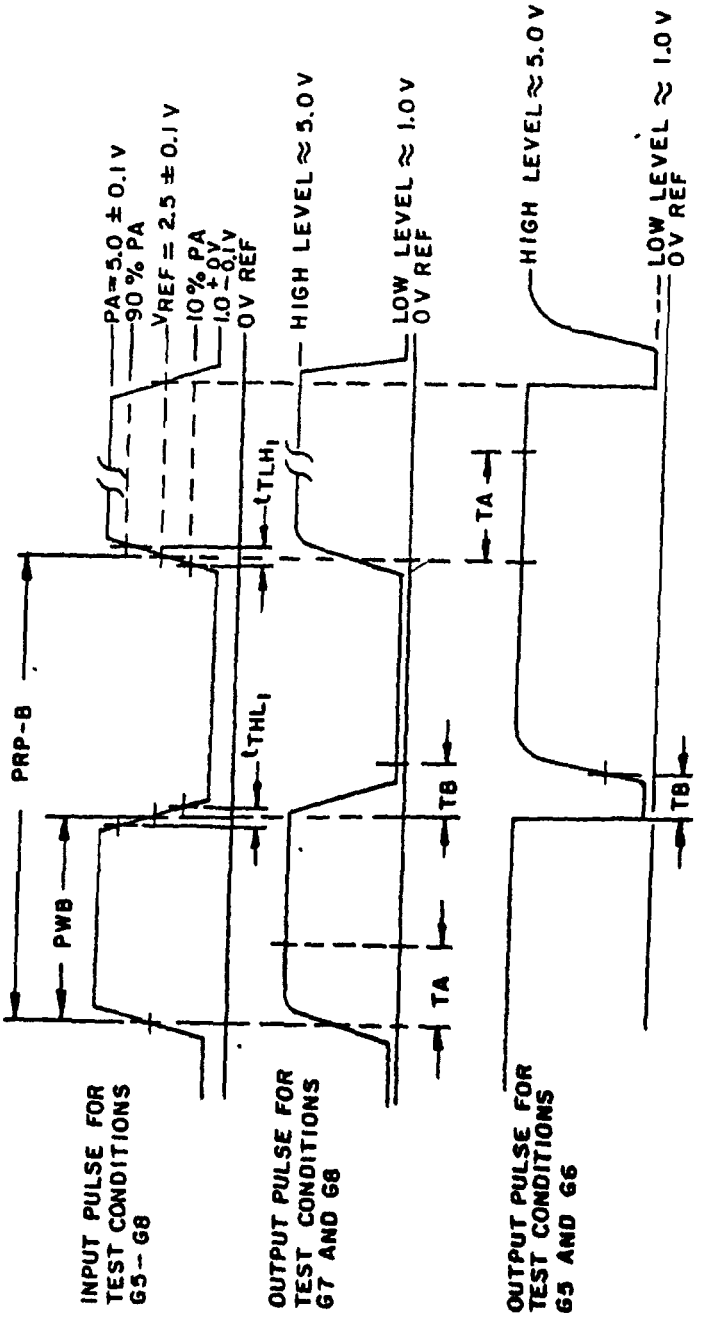
FIGURE 12. Power fail/restart test circuit for table VIII.

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* FIGURE 13. Waveforms for power fail/restart voltage check for table VIII.

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* FIGURE 13. Waveforms for power fail/restart voltage check for table VIII - Continued.

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NOTES:
1. Input pulse parameters (3 input pulses):
   PRP-A = 10 Hz ±5% (100 ms).
   PWA = 50 ±5 ms.
   $t_{TLH_1} = t_{THL_1} = \leq 200$ ns.
   PA = (as shown on each of 3 input pulses).
   PWB = 200 ±20 ms (5 Hz).
   PRP-B = 400 ±40 ms.
2. For test conditions G4 and G8, measure the voltage of the output waveform 30 ms (TA = 30 ms) after the rising edge of the input pulse and the readings should be within the limits specified in table VIII.
3. For test condition G2, measure the voltage of the output waveform 30 ms (TA = 30 ms) after the falling edge of the input pulse and the readings should be within the limits specified in table VIII.
4. For test condition G3, measure the voltage of the output waveform 500 µs (TB = 500 µs) after the falling edge of the input pulse and the readings should be within the limits specified in table VIII.
5. For test condition G5, measure the voltage of the output waveform 2.0 ms (TB = 2.0 ms) after the falling edge of the input pulse and the readings should be within the limits specified in table VIII.
6. For test condition G1, measure the voltage of the output waveform 500 µs (TB = 500 µs) after the rising edge of the input pulse and the readings should be within the limits specified in table VIII.
7. For test condition G7, measure the voltage of the output waveform 3.0 ms (TB = 3.0 ms) after the falling edge of the input pulse and the readings should be within the limits specified in table VIII.
8. For test condition G6, measure the voltage of the output waveform 150 ms (TA = 150 ms) after the rising edge of the input and the readings should be within the limits specified in table VIII.

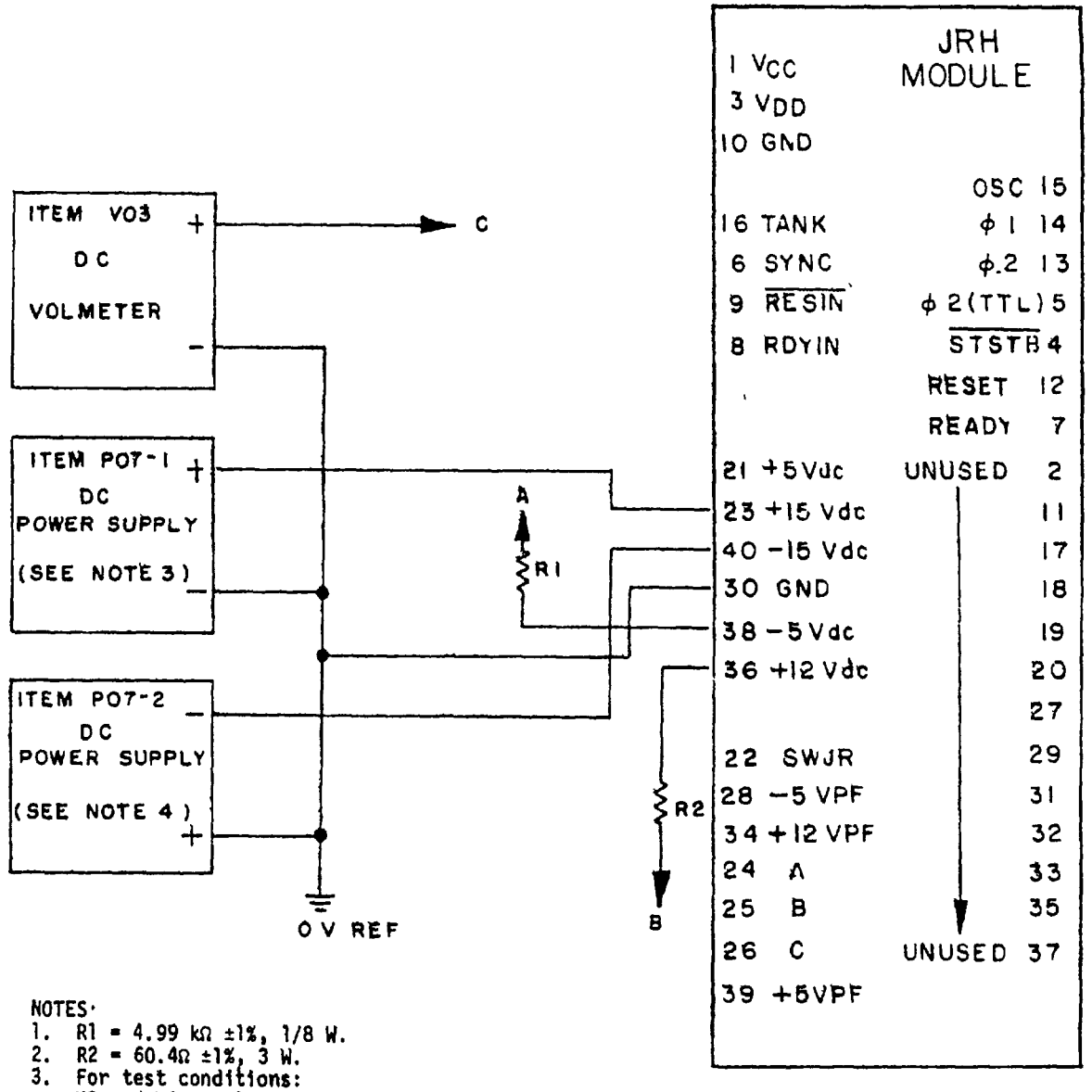
FIGURE 13. Waveforms for power fail/restart voltage check for table VIII - Continued.

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TABLE IX. Regulator output voltage test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	-5 V Regulator (unloaded)	Open	Open	38	-4.85	-5.35	V	14
H2	-5 V Regulator (loaded)	30	Open	38	-4.85	-5.35		
H3	12 V Regulator (unloaded)	Open	Open	36	11.4	12.6		
H4	12 V Regulator (loaded)	Open	30	36	11.4	12.6		
H5	-5 V Regulator (loaded)	30	Open	38	-4.85	-5.35		
H6	12 V Regulator (loaded)	Open	30	36	11.4	12.6	V	14

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NOTES:
1. R1 = 4.99 kΩ ±1%, 1/8 W.
2. R2 = 60.4Ω ±1%, 3 W.
3. For test conditions:
   H3 and H4 set item PO7-1 voltage = +15.0 V ±1%.
* H6 set item PO7-1 voltage = +16.5 V $^{+0}_{-1}$ %.
4. For test conditions:
   H1 and H2 set item PO7-2 voltage = -15.0 V ±1%.
* H5 set item PO7-2 voltage = -16.5 V $^{+0}_{-1}$ %.

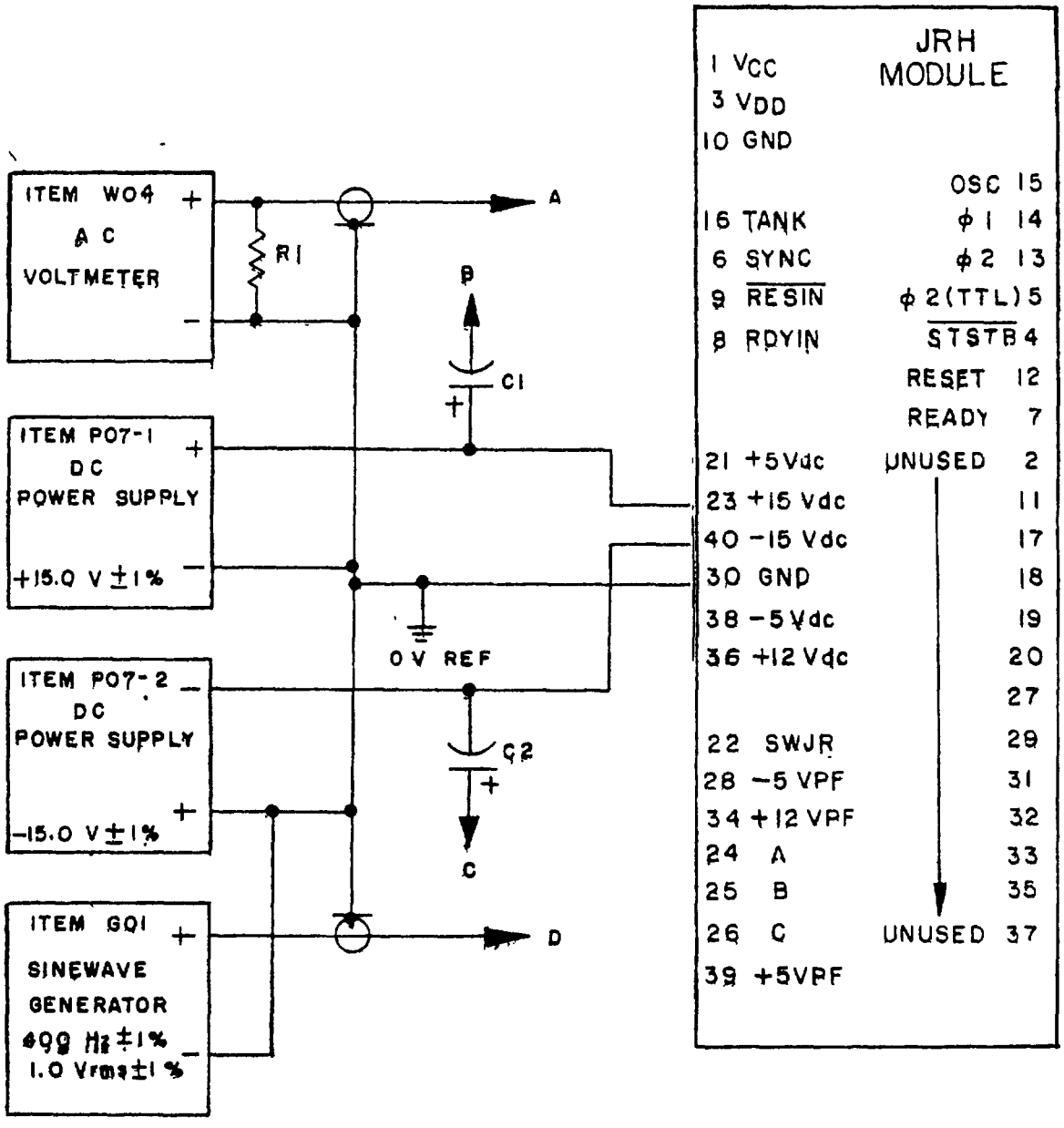
FIGURE 14. Regulator output voltage (-5 V and +12 V) test circuit for table IX.

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TABLE X. Regulator ripple rejection test.

Test condition	Parameter		Connect				Initial limits .25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	-5 V	Ripple rejection	38	Open	D	C	0	10	mVrms	15
J2	+12 V	Ripple rejection	36	D	Open	B	0	24	mVrms	15

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. C1, C2 = 20 µF ±20%.

FIGURE 15. Regulator ripple rejection test circuit for table X.

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TABLE XI. Output leakage current.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
K	A Output leakage current	0	500	nA	16

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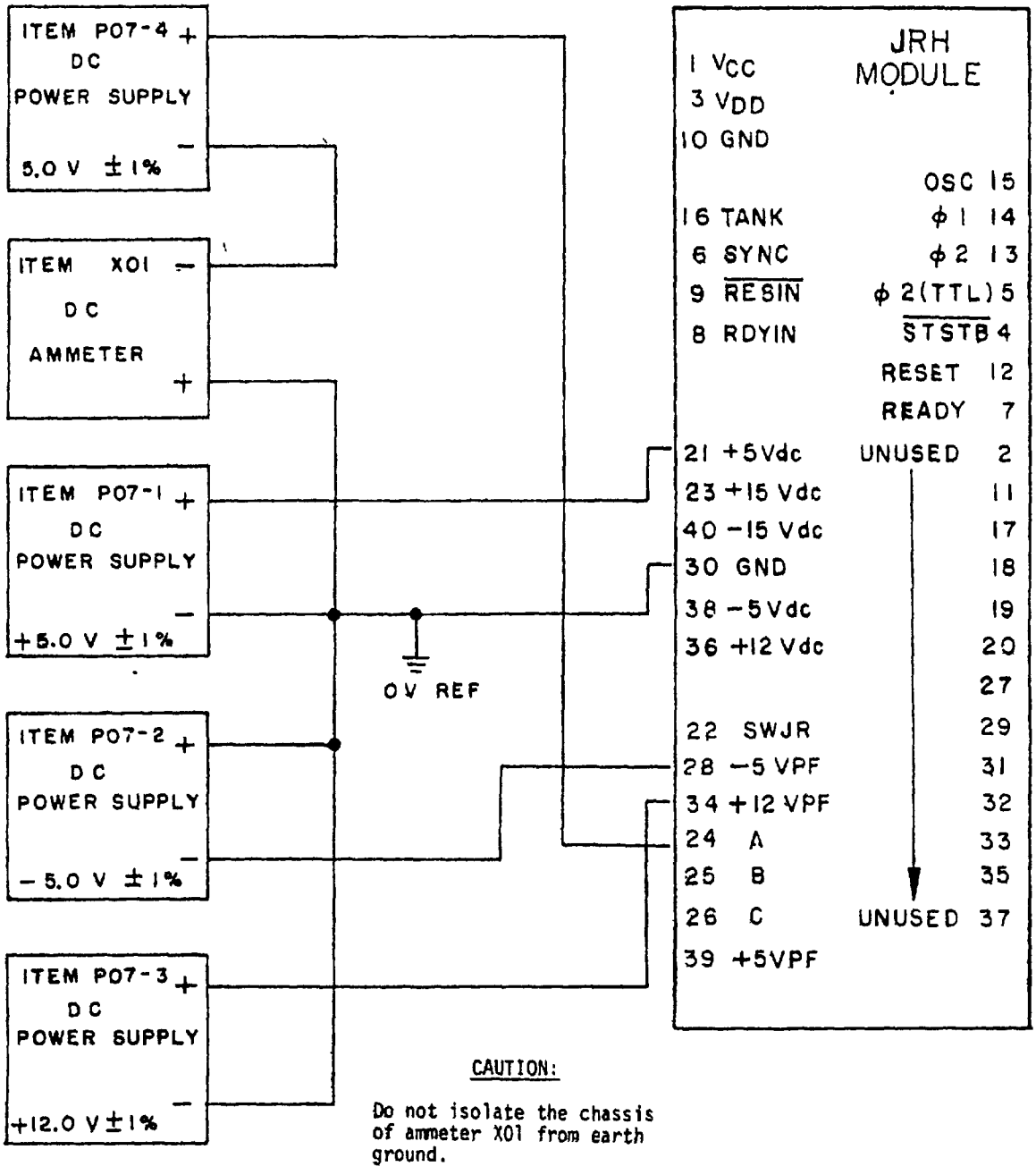


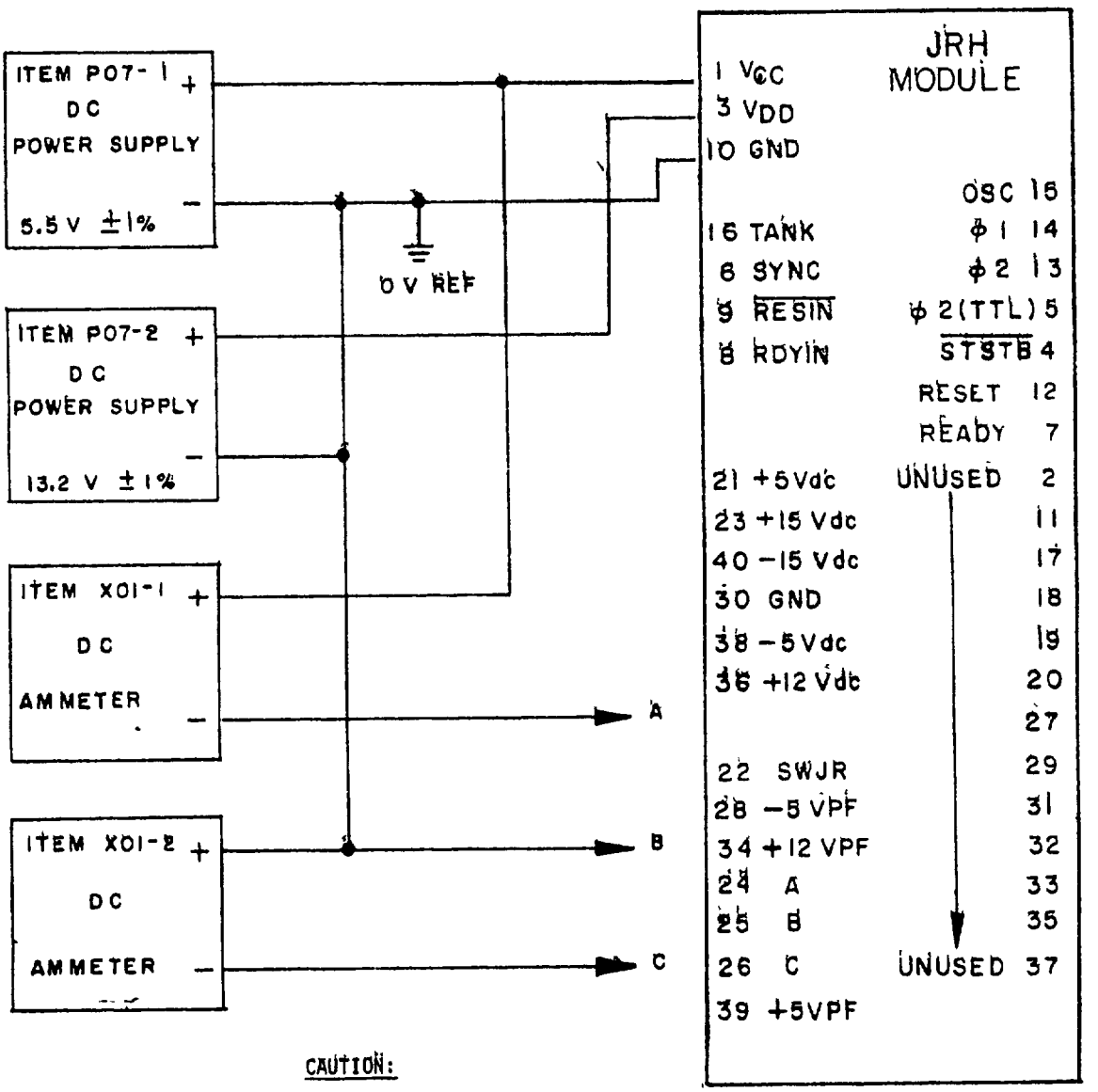
FIGURE 16. Output leakage current test circuit for table XI.

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TABLE XII. Input current (low-level and high-level) test.

	Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	Min	Max		
*	L1	Sync	Input current (low level)	Open	Open	6	-1	-270	µA	17
*	L2	$\overline{\text{Resin}}$	Input current (low level)	Open	Open	9	-1	-270		
*	L3	RDYIN	Input current (low level)	Open	Open	8	-1	-270		
	M1	Sync	Input current (high level)	6	9,8	Open	0	10		
	M2	$\overline{\text{Resin}}$	Input current (high level)	9	6,8	Open	0	10		
	M3	RDYIN	Input current (high level)	8	6,9	Open	0	10	µA	17

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Test conditions L1-L3 and M1-M3 are not run simultaneously. Therefore, only one ammeter is needed.

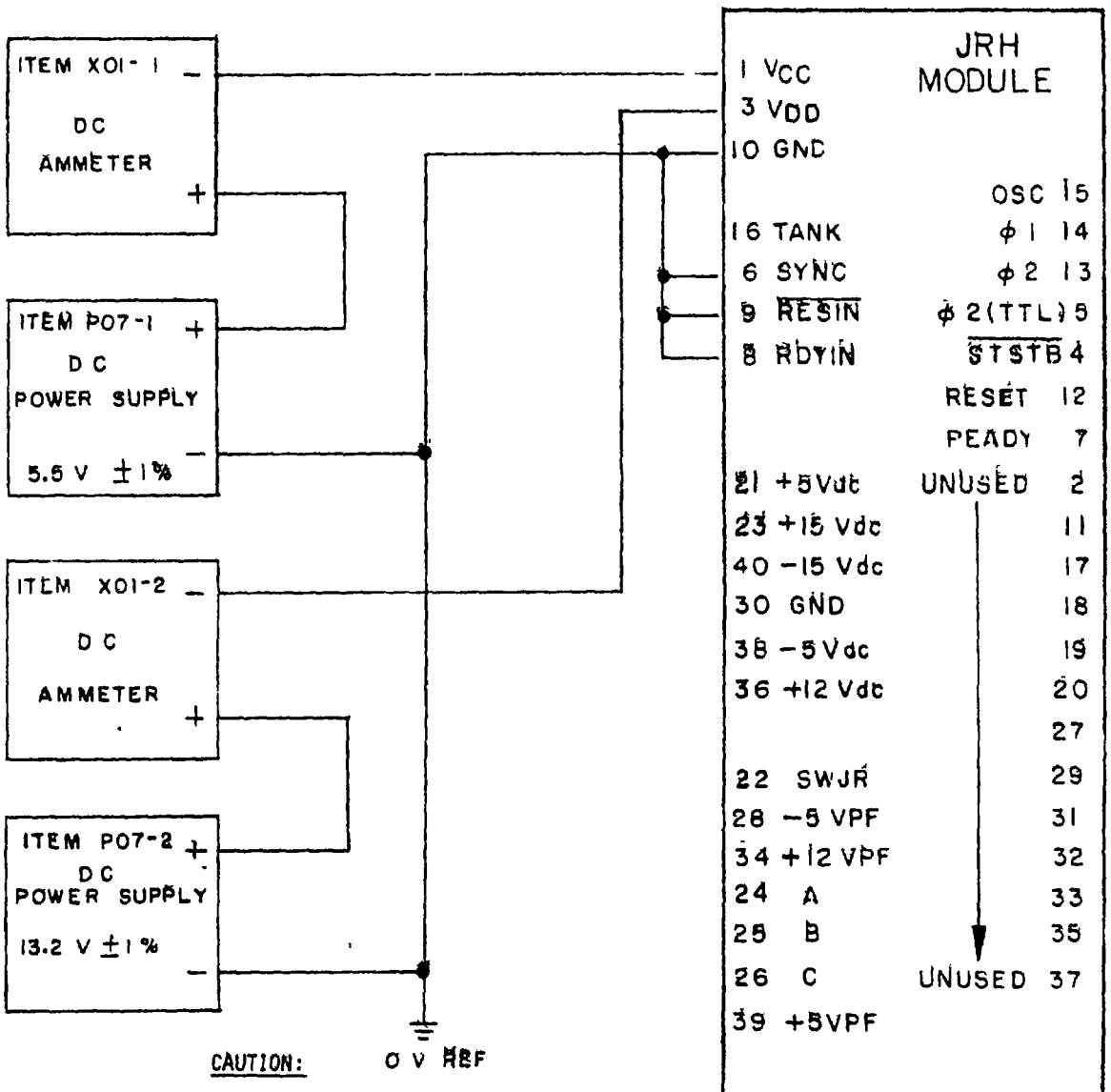
FIGURE 17. Input current test circuit for table XII.

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TABLE XIII. V_{DD} and V_{CC} power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
N1	V_{DD} Power supply current	0.01	12.0	mA	18
N2	V_{CC} Power supply current	0.01	115	mA	18

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Test conditions N1 and N2 are not run simultaneously. Therefore, only one ammeter is needed.

FIGURE 18. V_{DD} and V_{CC} power supply current test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits ·25° ±5°C Ambient Min	Initial limits ·25° ±5°C Ambient Max	Unit	Test figure
P1	ϕ1	Transition time, t_{TLH} (C_L = 50 pF)	6,8,9	Open	Open	14	Open	1	25	ns	19,20
P2	ϕ2					13			25		
P3	OSC					15			15		
P4	ϕ2(TTL)		6,8,9			5	Open				19,20
P5	$\overline{\text{STSTB}}$		8	Open		4	6,9				19,21
P6	Reset		8	9		12	6				19,21
P7	Ready	Transition time, t_{TLH} (C_L = 50 pF)	6,9	8	Open	7	Open	1	15	ns	19,21

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TABLE XV. Transition time (t_{THL}) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits ·25° ±5°C Ambient Min	Initial limits ·25° ±5°C Ambient Max	Unit	Test figure
Q1	Φ1	Transition time, t_{THL} (C_L = 50 pF)	6,8,9	Open	Open	14	Open	1	25	ns	19,20
Q2	Φ2					13			25		
Q3	OSC					15			15		
Q4	Φ2(TTL)		6,8,9	Open		5	Open				19,20
Q5	$\overline{STSTB}$		8	Open		4	6,9				19,21
Q6	Reset		8	9		12	6				19,21
Q7	Ready	Transition time, t_{THL} (C_L = 50 pF)	6,9	8	Open	7	Open	1	15	ns	19,21

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TABLE XVI. Output dynamic tests.

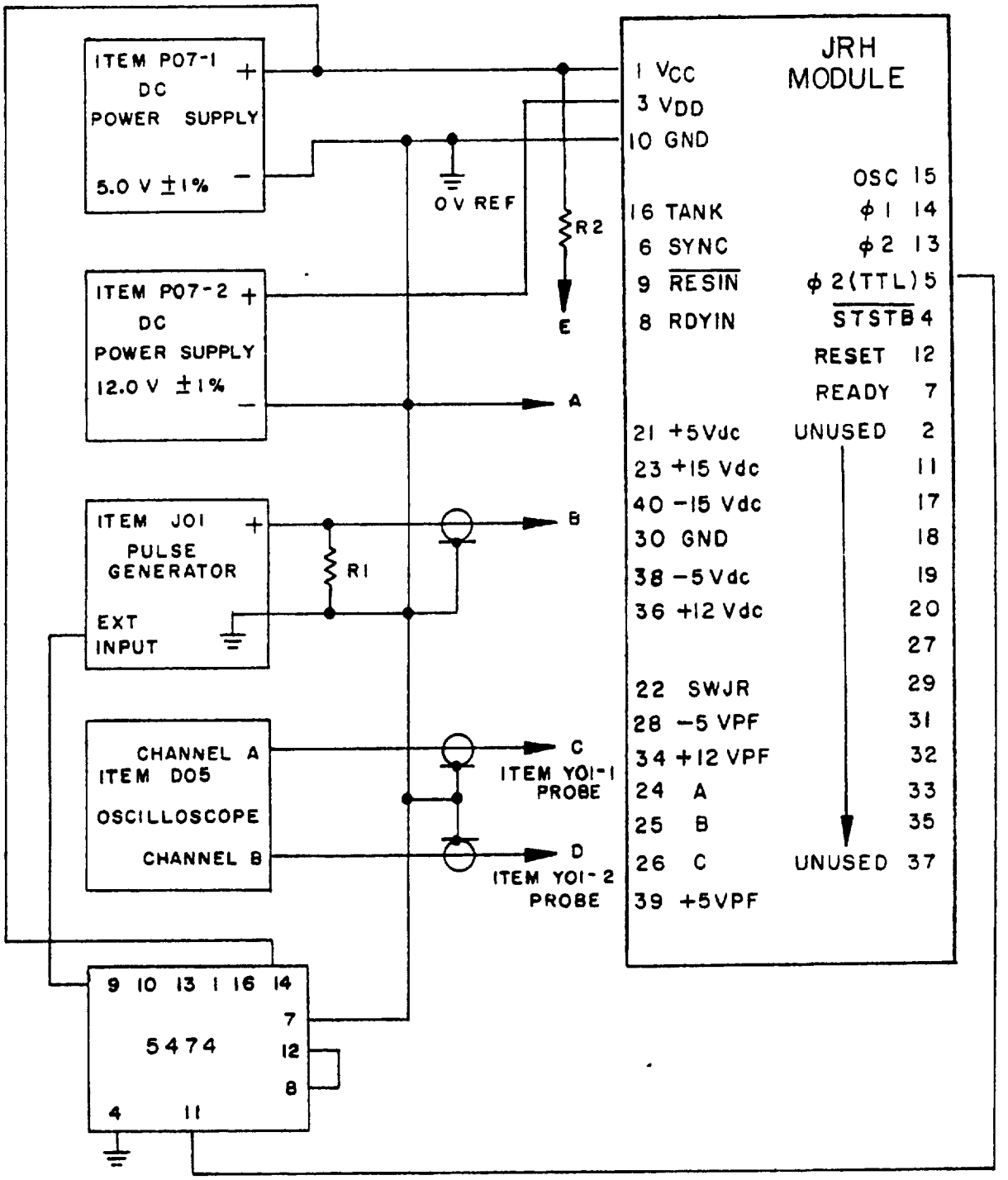
	Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
	R	$t_{\phi 1}$ $\phi 1$ pulse width check	6,8,9	Open	Open	14	Open	90	110	ns	19,20
	S	$t_{\phi 2}$ $\phi 2$ pulse width check			Open	13		231	275		
*	T	t_{D1} $\phi 1$ to $\phi 2$ delay			14	13		0	20		
	U	t_{D2} $\phi 2$ to $\phi 1$ delay			13	14		85	115		
	V	t_{D3} $\phi 1$ to $\phi 2$ delay			14	13		110	150		
	W	t_{DLH} $\phi 2$ to $\phi 2$ (TTL) delay			13	5		-5	+15		
	X	t_{DHL} $\phi 2$ to $\phi 2$ (TTL) delay			13	5		-5	+15		
*	Y	T_{CY} $\phi 1$ pulse period check	6,8,9		14	Open	Open	495	505		19,20
	Z	t_{DSS} $\phi 2$ to $\overline{STSTB}$ delay	8	Open	13	4	6,9	304	334	ns	19,21

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TABLE XVI. Output dynamic tests - Continued.

	Test condition	Parameter		Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
*	AA	t_{PW}	$\overline{STSTB}$ pulse width check	8	Open	Open	4	6,9	32	60	ns	19,21
	BB	t_{DR1}	Ready to Ø2 delay (t_{PHL})	Open	8	7	13	6,9	200	230		
	CC	t_{DR2}	Ready to Ø2 delay (t_{PHL})	Open	8	7	13	6,9	200	230		
*	DD	t_{DR3}	Reset to Ø2 delay (t_{PHL})	8	9	12	13	6	200	230	ns	19,21

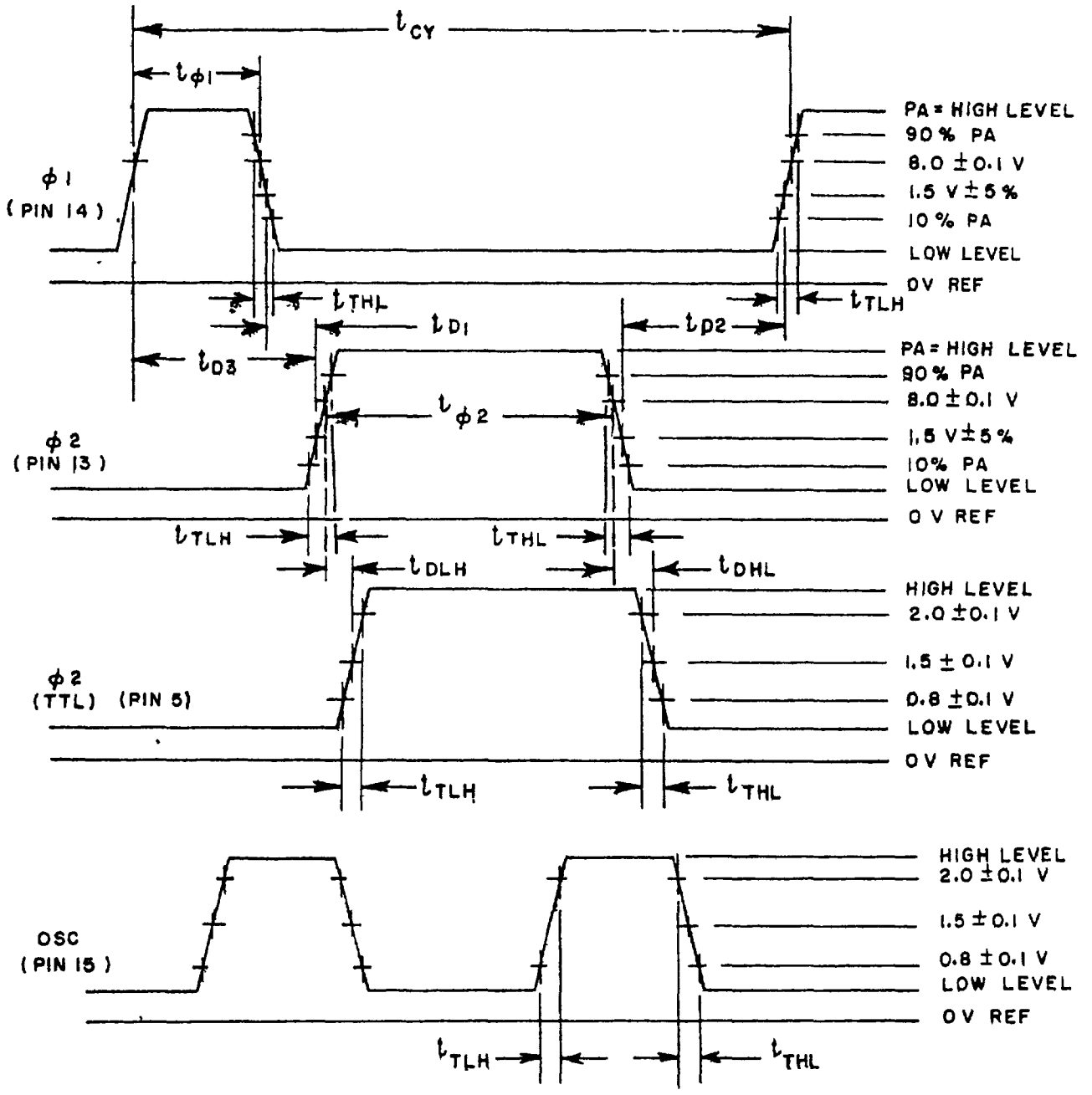
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load all outputs of circuit under test with load as specified on figure 11.
3. R2 = 10.0 kΩ ±1%, 1/8 W.

FIGURE 19. Transition time and output dynamic test circuit for tables XIV, XV, and XVI.

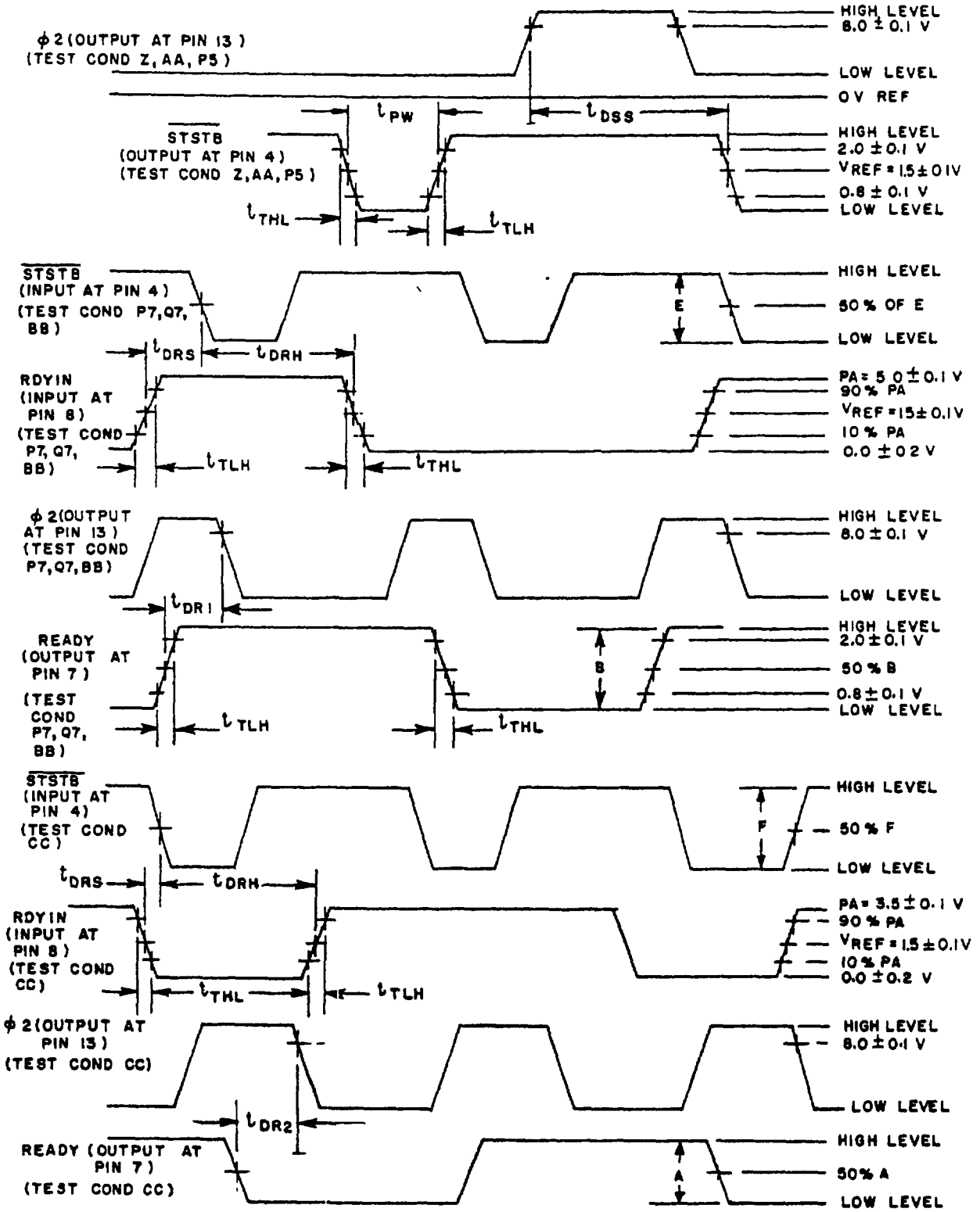
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NOTE: Before making the specified measurements, verify that the low level output is ≤ 0.45 V and that the high level output is ≥ 4.5 V for OSC and $\phi 2$ (TTL) outputs and ≥ 10.8 V for $\phi 1$ and $\phi 2$ outputs.

★ FIGURE 20. Transition time and output dynamic test circuit output waveforms for tables XIV, XV, and XVI.

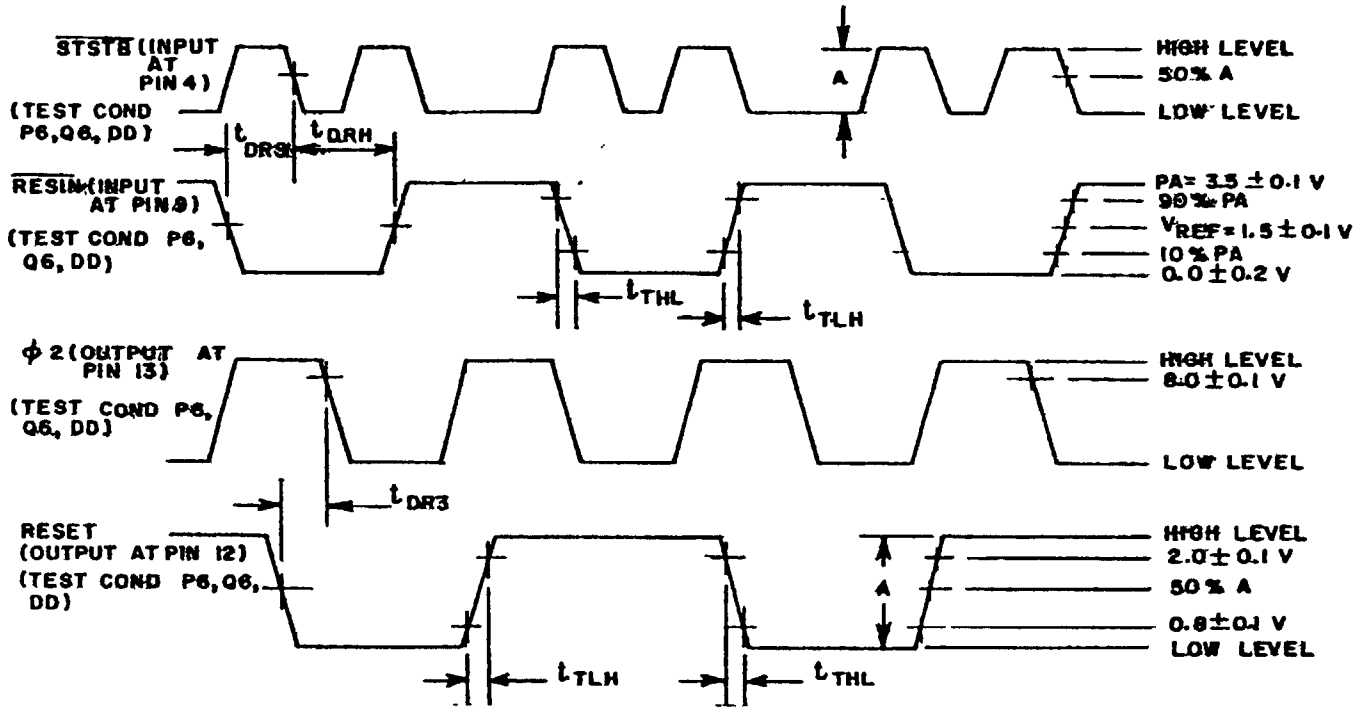
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* NOTE Before making the specified measurements, verify that the low level output ≤0.45 V and that the high level output is ≥ 4.5 V for OSC and φ2 (TTL) outputs and ≥10.8 V for φ1 and φ2 outputs.

FIGURE 21. Transition time and output dynamic test circuit waveforms for tables XIV, XV and XVI.

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* NOTES:

1. Input pulse conditions:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{DRS} = 171\ ^{+4}_{-0}$ ns.
   $t_{DRH} = 221\ ^{+4}_{-0}$ ns.
2. Before making the specified measurements, verify that the low level output is $\leq$0.45 V and that the high level output is $\geq$4.5 V for OSC and ϕ2 (TTL) outputs and $\geq$10.8 V for ϕ1 and ϕ2 outputs.

FIGURE 21. Transition time and output dynamic test circuit waveforms for tables XIV, XV, and XVI - Continued.

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TABLE XVII. Frequency check test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
EE1	Frequency check, 18 MHz	3	15	Open	17.9	18.1	MHz	22
EE2	Frequency check, 2 MHz	Open	5	3	1.98	2.02	MHz	22

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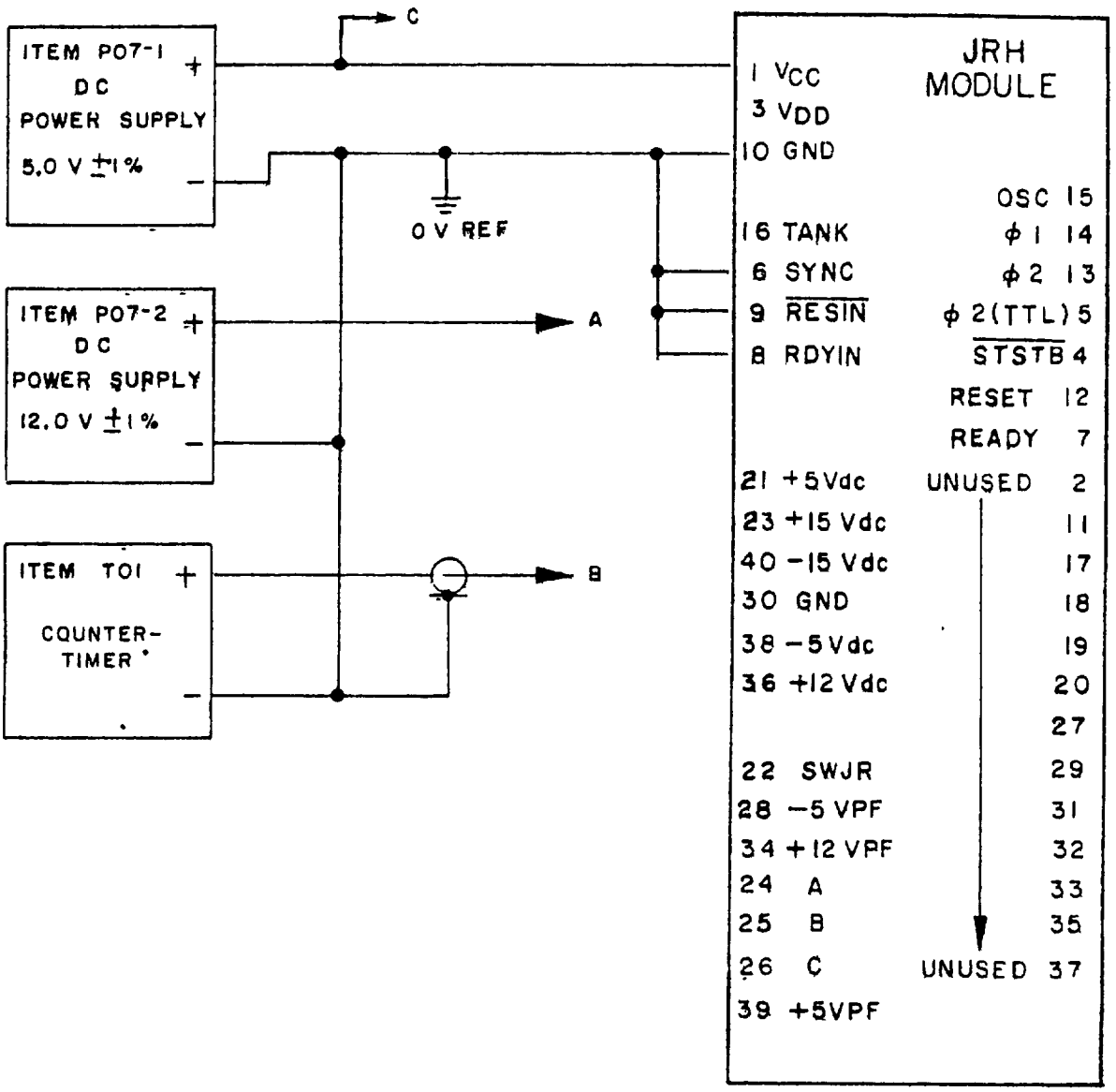


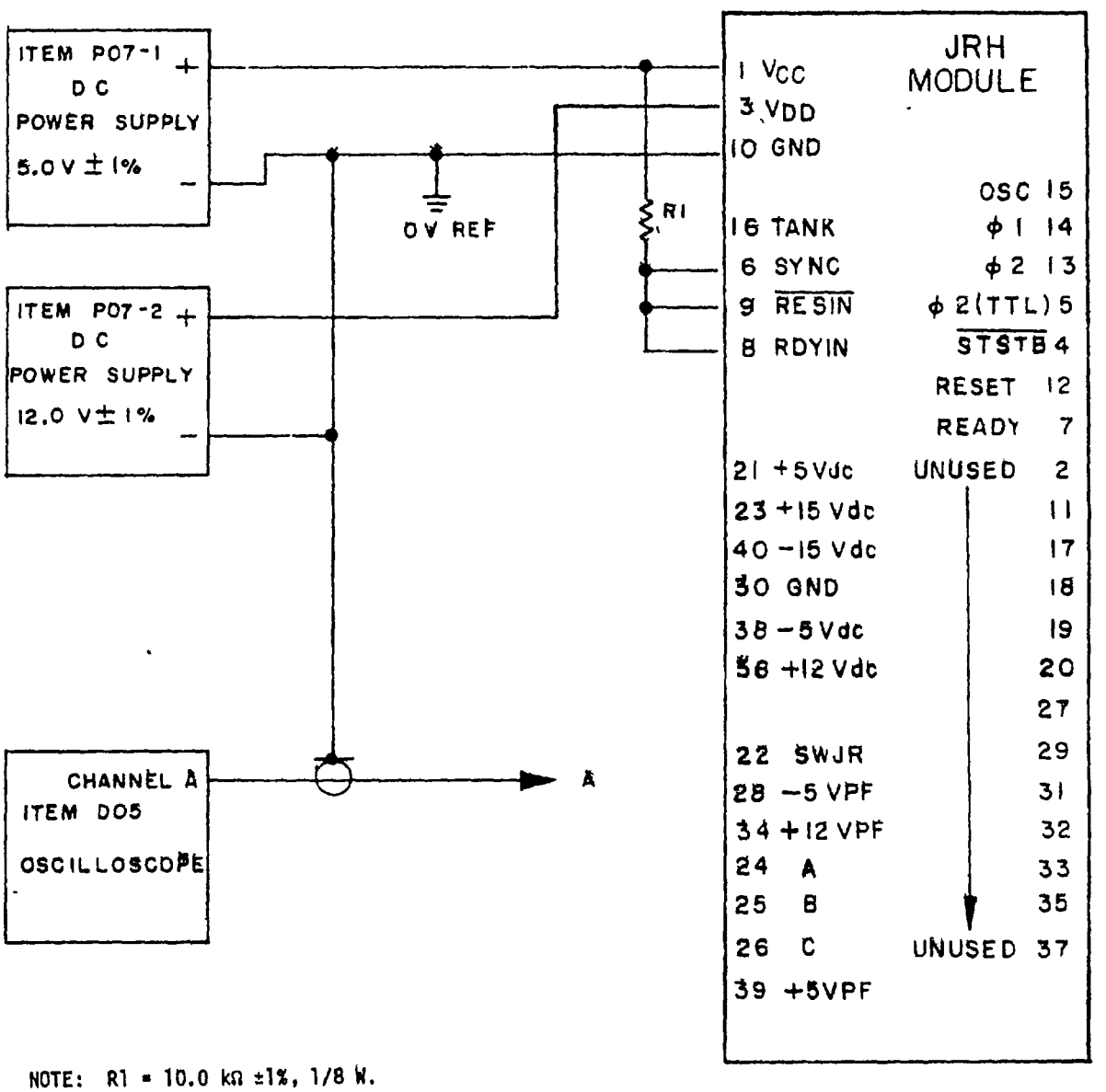
FIGURE 22. Frequency check test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
FF1	V_{CC}	Power supply transient amplitude test	1	0	400	mV	23,24
FF2	V_{DD}	Power supply transient amplitude test	3	0	400	mV	23,24

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NOTE: R1 = 10.0 kΩ ±1%, 1/8 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVIII.

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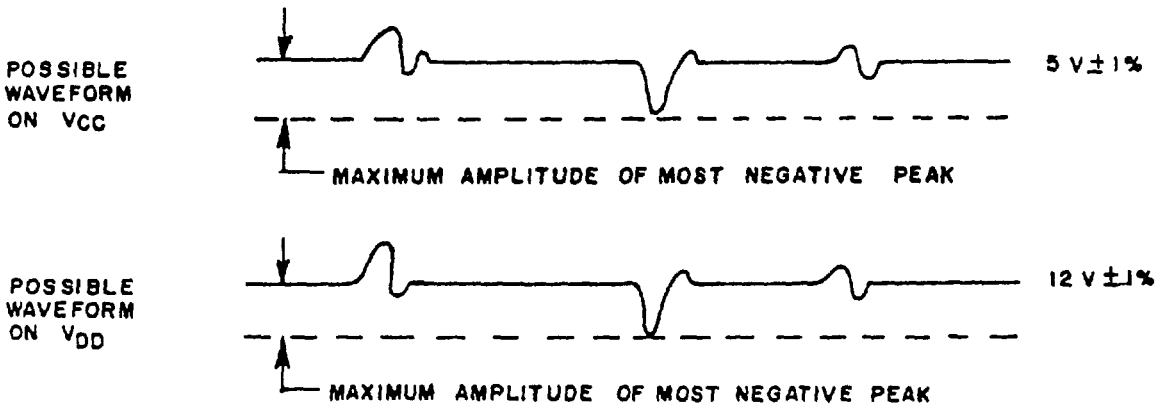


FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (low-level) test.

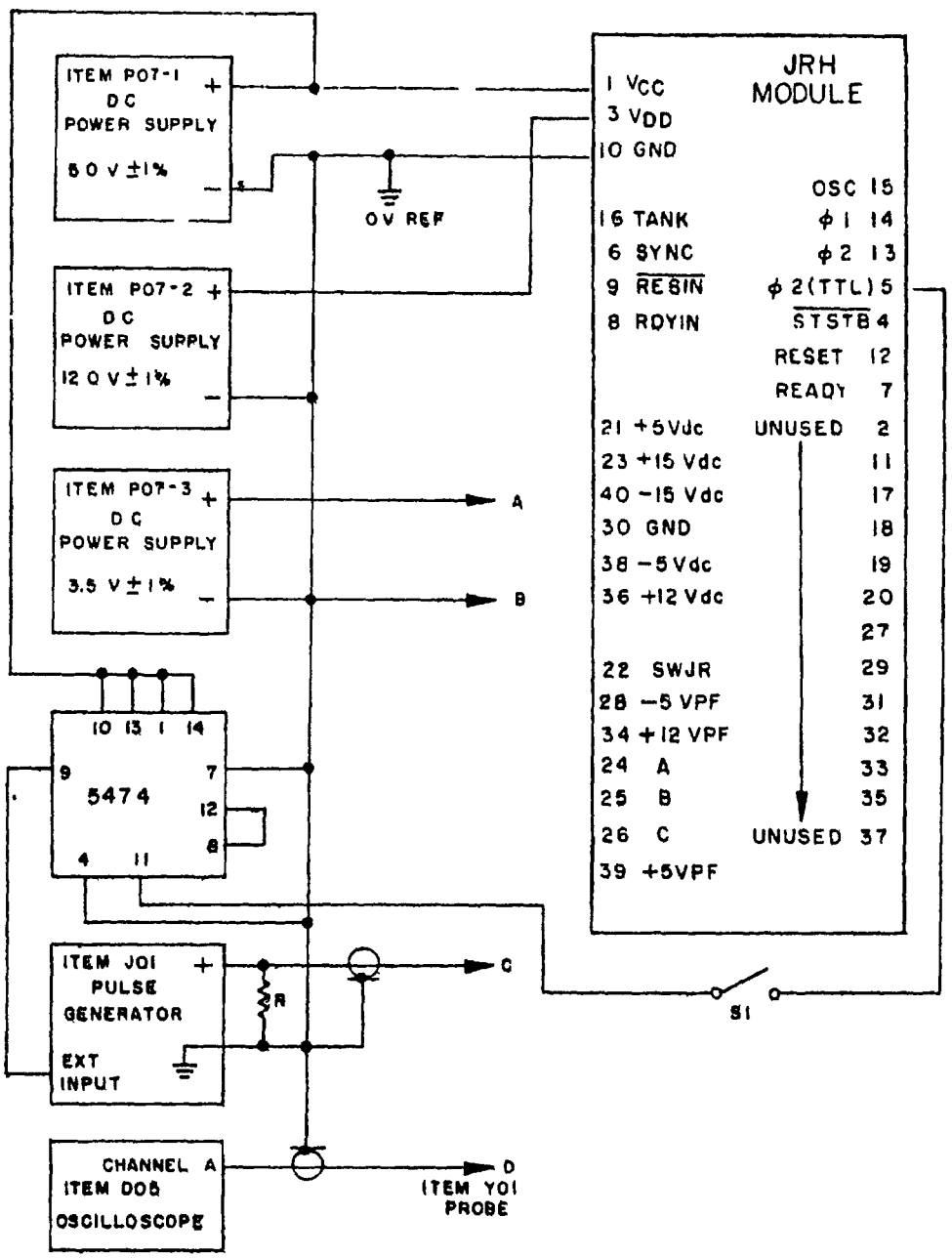
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
GG1	$\overline{\text{STSTB}}$	Output crosstalk, low level	Open	6,9	8	4	0	500	mV	25,26
GG2	Reset		6,9	Open	8	12		500		
GG3	Ready		6	8	9	7		500		
GG4	OSC		6	Open	8,9	15		850		
GG5	ϕ1		6	Open	8,9	14		1,000		
GG6	ϕ2		6	Open	8,9	13		1,000		
GG7	ϕ2(TTL)	Output crosstalk, low level	6	8,9	Open	5	0	850	mV	25,26

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* TABLE XX. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
HH1	$\overline{\text{STSTB}}$	Output crosstalk, high level	9	6	8	4	0	400	mV	25,26
HH2	Reset		6	9	8	12				
HH3	Ready		6,8	Open	9	7				
HH4	OSC		6		8,9	15				
HH5	ϕ1				8,9	14				
HH6	ϕ2			Open	8,9	13				
HH7	ϕ2(TTL)	Output crosstalk, high level	6	8,9	Open	5	0	400	mV	25,26

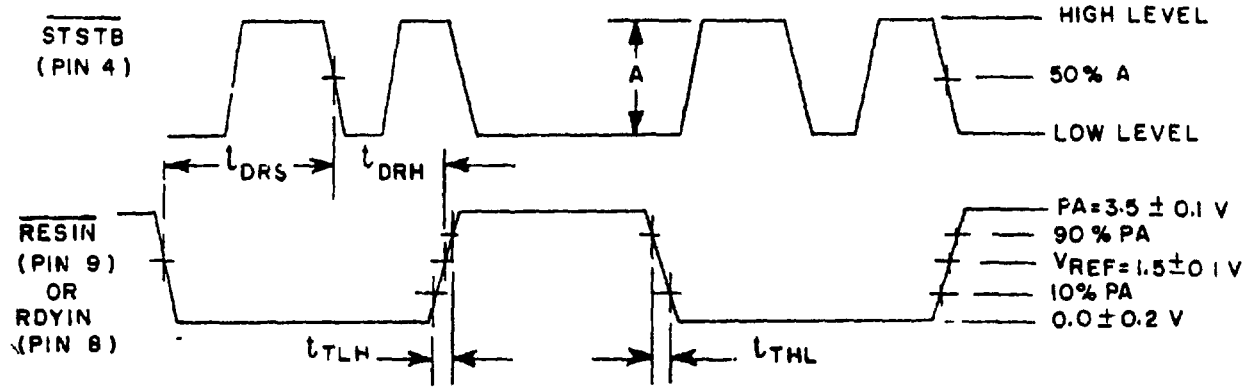
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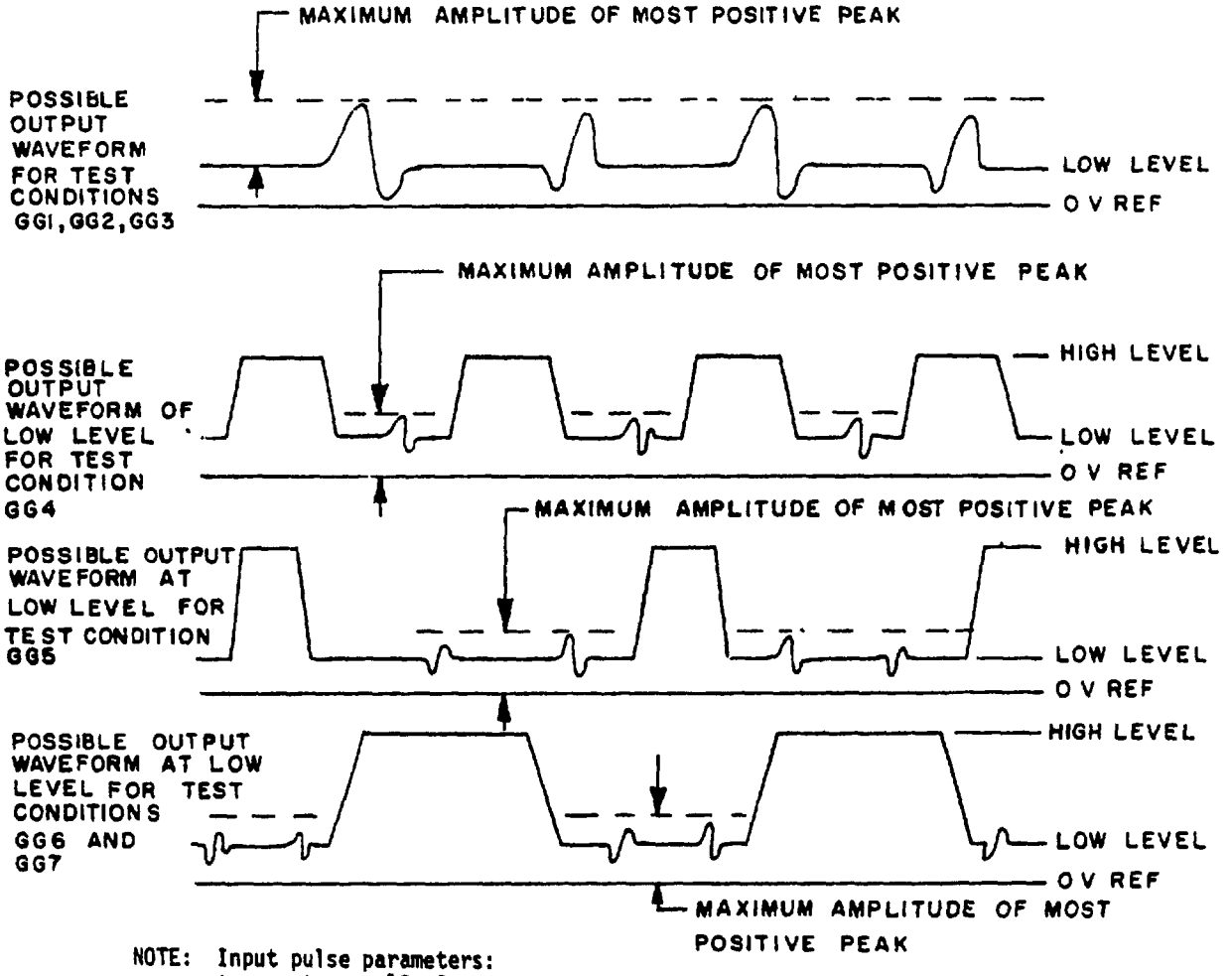
NOTES:
1. R = 51.1Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.
3. Close switch S1 for test conditions GG1-GG6 and HH1-HH6 and open S1 for test conditions GG7 and HH7.
4. For test condition HH1, connect pin 6 to A until input waveforms are setup then connect pin 6 to B.
* 5. Use the chopper for test conditions GG4-GG7.

FIGURE 25. Output crosstalk (low-level and high-level) test circuit for tables XIX and XX.

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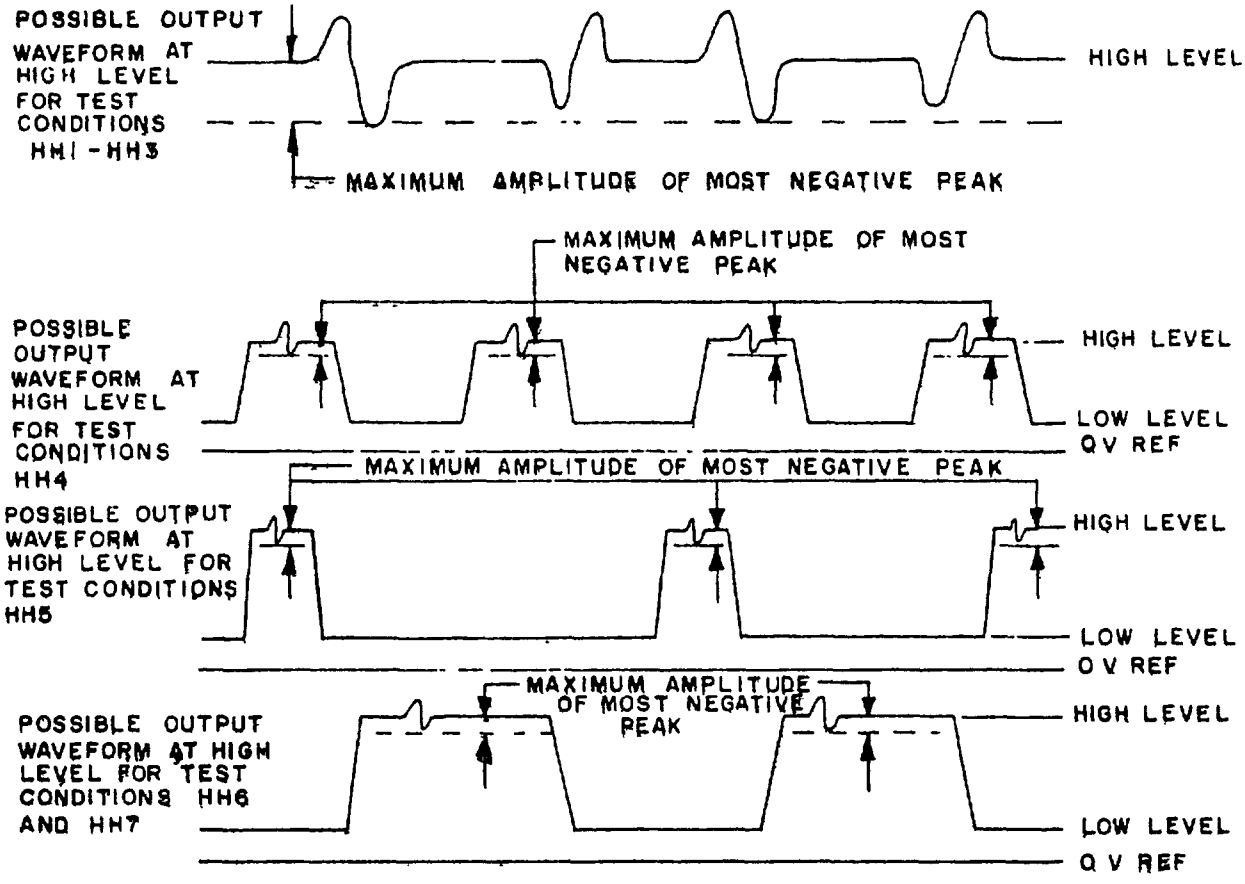
THE ABOVE INPUT WAVEFORMS ARE FOR TEST CONDITIONS GG1-GG6 AND HH1-HH6



NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
$t_{DRS} = 200 \pm 10$ ns.
$t_{DRH} = 250 \pm 10$ ns.

FIGURE 26. Output crosstalk test circuit waveforms for tables XIX and XX.

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* FIGURE 26. Output crosstalk test circuit waveforms for tables XIX and XX - Continued.

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TABLE XXI. Power supply filter capacitance test.

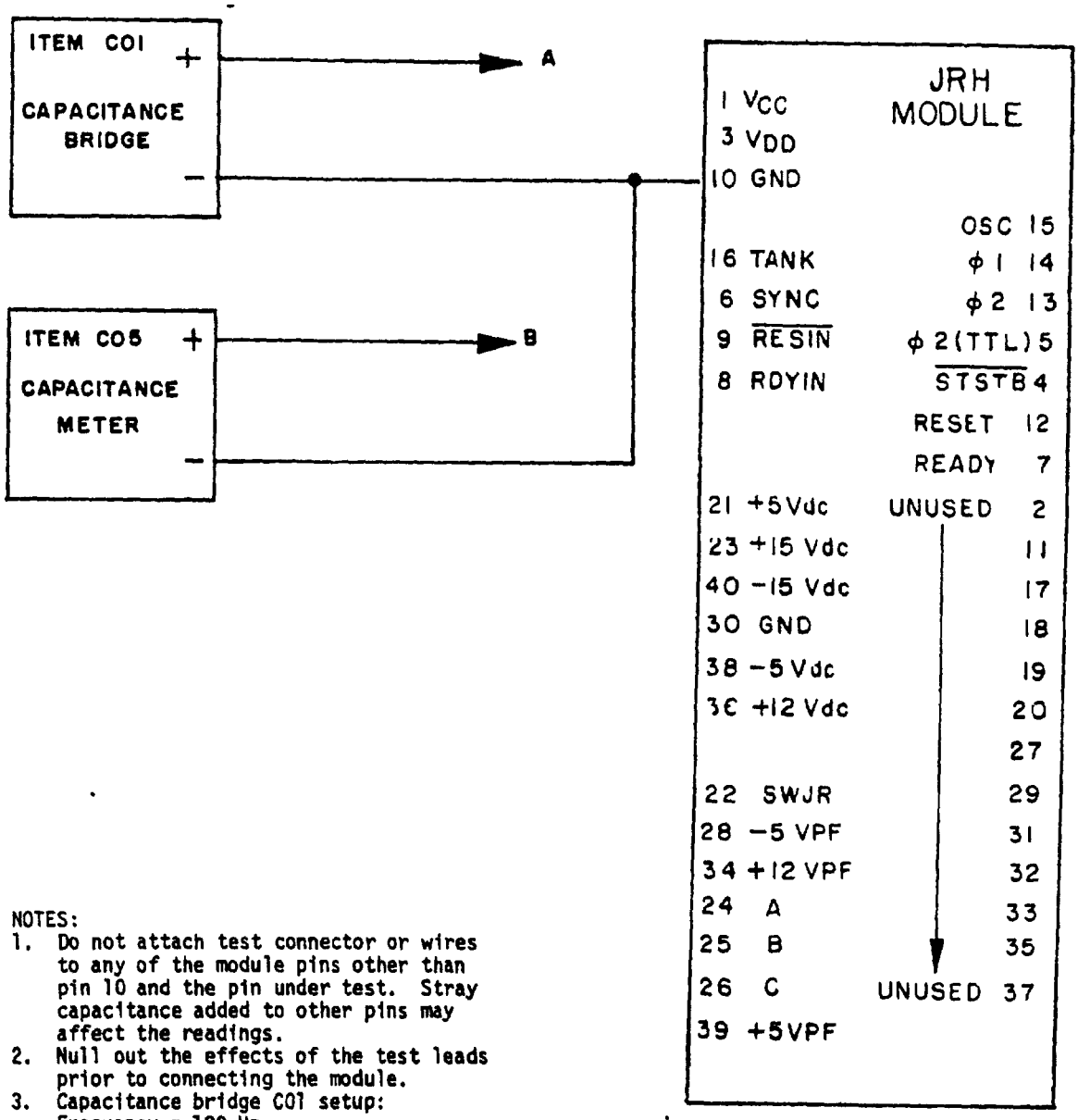
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
* JJ1	V_{CC}	Power supply filter capacitance	1	Open	0.71	1.31	µF	27
* JJ2	V_{DD}	Power supply filter capitance	3	Open	0.71	1.31	µF	27

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TABLE XXII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
KK1	Sync	Input capacitance	Open	6	2	15	pF	27
KK2	$\overline{\text{Resin}}$	Input capacitance	Open	9	2	15	pF	27
KK3	RDYIN	Input capacitance	Open	8	2	15	pF	27

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.
3. Capacitance bridge C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 27. Power supply filter and input capacitance test circuit for tables XXI and XXII.

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